0902.0431 — formula report

0 inline formulas, 57 display equations, 0 tables, 0 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ0515 **[conf 0.002]**	085	0.002	`\begin{aligned} & \lambda_{0}-\lambda_{1}=1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\ & \lambda_{0}-\lambda_{2}=1 \quad 1 \quad 0 \quad 0 \quad 0 \quad \lambda_{0}+ \lambda_{1}=1 \quad 2 \\ & \lambda_{0}-\lambda_{3}=1 \quad 1 \quad 1 \quad 0 \quad 2 \end{aligned} \quad \frac{1}{1} \text { = } \begin{array}{llllllll} 1 & 1 & 1 & 0 & 0 & \lambda_{0}+\lambda_{3}=1 & 1 & 2 \\ \lambda_{1}-\lambda_{2} & =0 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}`	$\begin{aligned} & \lambda_{0}-\lambda_{1}=1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\ & \lambda_{0}-\lambda_{2}=1 \quad 1 \quad 0 \quad 0 \quad 0 \quad \lambda_{0}+ \lambda_{1}=1 \quad 2 \\ & \lambda_{0}-\lambda_{3}=1 \quad 1 \quad 1 \quad 0 \quad 2 \end{aligned} \quad \frac{1}{1} \text { = } \begin{array}{llllllll} 1 & 1 & 1 & 0 & 0 & \lambda_{0}+\lambda_{3}=1 & 1 & 2 \\ \lambda_{1}-\lambda_{2} & =0 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}$	$\begin{array}{lllllll} \lambda_0-\lambda_1 & = 1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0-\lambda_2 & = 1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0-\lambda_3 & = 1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_1-\lambda_2 & = 0 & 1 & 0 & 0 & 0 & 0 \\ \lambda_1-\lambda_3 & = 0 & 1 & 1 & 0 & 0 & 0 \\ \lambda_2-\lambda_3 & = 0 & 0 & 1 & 0 & 0 & 0 \end{array} \quad \begin{array}{lllllll} \lambda_0+\lambda_1 & = 1 & 2 & 2 & 1 & 1 & 1 \\ \lambda_0+\lambda_2 & = 1 & 1 & 2 & 1 & 1 & 1 \\ \lambda_0+\lambda_3 & = 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1+\lambda_2 & = 0 & 1 & 2 & 1 & 1 & 1 \\ \lambda_1+\lambda_3 & = 0 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2+\lambda_3 & = 0 & 0 & 1 & 1 & 1 & 1 \end{array}$

Conf. MathPix confidence: $\geq$0.9 0.5–0.9 <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_ EQ0517 **[conf 0.002]**	085	■ 0.002	`\begin{aligned} & \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}+\lambda_{2}-\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=0 \quad 0 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}\left(\lambda_{0}+\lambda_{1}+\lambda_{2}-\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=1 \quad 2 \quad 3 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}+\lambda_{2}+\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=0 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2}\left(\quad \lambda_{0}-\lambda_{1}+\lambda_{2}+\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\quad \lambda_{0}+\lambda_{1}-\lambda_{2}+\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=1 \quad 2 \quad 2 \quad 1 \\ & \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}-\lambda_{2}+\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=0 \quad 0,0 \quad 0 \\ & \frac{1}{2}\left(\quad \lambda_{0}-\lambda_{1}-\lambda_{2}-\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}-\lambda_{2}-\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=0 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2}\left(\quad \lambda_{0}-\lambda_{1}+\lambda_{2}-\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=1 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2}\left(\quad \lambda_{0}-\lambda_{1}-\lambda_{2}+\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=1 \quad 1 \quad 1 \end{aligned} \quad \frac{1}{1} \quad 1 \quad 0`	$\begin{aligned} & \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}+\lambda_{2}-\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=0 \quad 0 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}\left(\lambda_{0}+\lambda_{1}+\lambda_{2}-\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=1 \quad 2 \quad 3 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}+\lambda_{2}+\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=0 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2}\left(\quad \lambda_{0}-\lambda_{1}+\lambda_{2}+\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\quad \lambda_{0}+\lambda_{1}-\lambda_{2}+\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=1 \quad 2 \quad 2 \quad 1 \\ & \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}-\lambda_{2}+\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=0 \quad 0,0 \quad 0 \\ & \frac{1}{2}\left(\quad \lambda_{0}-\lambda_{1}-\lambda_{2}-\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}-\lambda_{2}-\lambda_{3}\right)+\frac{1}{2}\left(\mu_{3}-\mu_{1}\right)=0 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2}\left(\quad \lambda_{0}-\lambda_{1}+\lambda_{2}-\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=1 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2}\left(\quad \lambda_{0}-\lambda_{1}-\lambda_{2}+\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=1 \quad 1 \quad 1 \end{aligned} \quad \frac{1}{1} \quad 1 \quad 0$	$\begin{aligned} \frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2-\lambda_3)+\frac{1}{2}(\mu_3-\mu_1) &= 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \\ \frac{1}{2}(\quad \lambda_0+\lambda_1+\lambda_2-\lambda_3)+\frac{1}{2}(\mu_3-\mu_1) &= 1 \quad 2 \quad 3 \quad 1 \quad 2 \quad 1 \\ \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}(\mu_3-\mu_1) &= 0 \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \\ \frac{1}{2}(\quad \lambda_0-\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}(\mu_3-\mu_1) &= 1 \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \\ \frac{1}{2}(\quad \lambda_0+\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}(\mu_3-\mu_1) &= 1 \quad 2 \quad 2 \quad 1 \quad 2 \quad 1 \\ \frac{1}{2}(-\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}(\mu_3-\mu_1) &= 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \\ \frac{1}{2}(\quad \lambda_0-\lambda_1-\lambda_2-\lambda_3)+\frac{1}{2}(\mu_3-\mu_1) &= 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \\ \frac{1}{2}(-\lambda_0+\lambda_1-\lambda_2-\lambda_3)+\frac{1}{2}(\mu_3-\mu_1) &= 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \\ \frac{1}{2}(\quad \lambda_0-\lambda_1+\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2) &= 1 \quad 1 \quad 2 \quad 1 \quad 1 \quad 0 \\ \frac{1}{2}(\quad \lambda_0-\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2) &= 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \frac{1}{2}(\quad \lambda_0+\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}(\mu_1-\mu_2) &= 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_ EQ1026 **[conf 0.018]**	171	0.018	`\left.\begin{array}{lllllllrllllll} \lambda_{0}-\lambda_{1}=2 & 3 & 4 & 3 & 2 & 1 & 0 & 2 & \lambda_{0}+\lambda_{1}=2 & 4 & 6 & 5 & 4 & 3 & 2 \end{array}\right], \begin{array}{llllllllll} \lambda_{0}-\lambda_{2}=2 & 3 & 4 & 3 & 2 & 1 & 1 & 2 & \lambda_{0}+\lambda_{2}=2 & 4 & 6 \\ \lambda_{0}-\lambda_{3}=2 & 3 & 4 & 3 & 2 & 2 & 1 & 2 & \lambda_{0}+\lambda_{3}=2 & 4 & 6 \\ \lambda_{1}-\lambda_{2}=0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 4 & 2 & 1 \\ \lambda_{1}-\lambda_{3}=0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & \lambda_{1}+\lambda_{2}=0 & 1 & 2 \\ \lambda_{2}-\lambda_{3}=0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 \\ \lambda_{2}+\lambda_{3}=0 & 1 & 2 & 2 & 2 & 1 & 0 & 1 \end{array}`	$\left.\begin{array}{lllllllrllllll} \lambda_{0}-\lambda_{1}=2 & 3 & 4 & 3 & 2 & 1 & 0 & 2 & \lambda_{0}+\lambda_{1}=2 & 4 & 6 & 5 & 4 & 3 & 2 \end{array}\right], \begin{array}{llllllllll} \lambda_{0}-\lambda_{2}=2 & 3 & 4 & 3 & 2 & 1 & 1 & 2 & \lambda_{0}+\lambda_{2}=2 & 4 & 6 \\ \lambda_{0}-\lambda_{3}=2 & 3 & 4 & 3 & 2 & 2 & 1 & 2 & \lambda_{0}+\lambda_{3}=2 & 4 & 6 \\ \lambda_{1}-\lambda_{2}=0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 4 & 2 & 1 \\ \lambda_{1}-\lambda_{3}=0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & \lambda_{1}+\lambda_{2}=0 & 1 & 2 \\ \lambda_{2}-\lambda_{3}=0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 \\ \lambda_{2}+\lambda_{3}=0 & 1 & 2 & 2 & 2 & 1 & 0 & 1 \end{array}$	$\begin{array}{llllllllllllllll} \lambda_0-\lambda_1=2 & 3 & 4 & 3 & 2 & 1 & 0 & 2 & \lambda_0+\lambda_1=2 & 4 & 6 & 5 & 4 & 3 & 2 & 3 \\ \lambda_0-\lambda_2=2 & 3 & 4 & 3 & 2 & 1 & 1 & 2 & \lambda_0+\lambda_2=2 & 4 & 6 & 5 & 4 & 3 & 1 & 3 \\ \lambda_0-\lambda_3=2 & 3 & 4 & 3 & 2 & 2 & 1 & 2 & \lambda_0+\lambda_3=2 & 4 & 6 & 5 & 4 & 2 & 1 & 3 \\ \lambda_1-\lambda_2=0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & \lambda_1+\lambda_2=0 & 1 & 2 & 2 & 2 & 2 & 1 & 1 \\ \lambda_1-\lambda_3=0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & \lambda_1+\lambda_3=0 & 1 & 2 & 2 & 2 & 1 & 1 & 1 \\ \lambda_2-\lambda_3=0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \lambda_2+\lambda_3=0 & 1 & 2 & 2 & 2 & 1 & 0 & 1 \end{array}$
0902.0431_ EQ0518 **[conf 0.021]**	086	0.021	`\begin{aligned} & \frac{1}{2}\left(\lambda_{0}+\lambda_{1}-\lambda_{2}-\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \quad c \\ & \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}-\lambda_{2}+\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ & \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}+\lambda_{2}-\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_{0}+\lambda_{1}+\lambda_{2}+\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=1 \quad 2 \quad 3 \quad 2 \\ & \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}+\lambda_{2}+\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=0 \quad 0 \end{aligned}`	$\begin{aligned} & \frac{1}{2}\left(\lambda_{0}+\lambda_{1}-\lambda_{2}-\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \quad c \\ & \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}-\lambda_{2}+\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ & \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}+\lambda_{2}-\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_{0}+\lambda_{1}+\lambda_{2}+\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=1 \quad 2 \quad 3 \quad 2 \\ & \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}+\lambda_{2}+\lambda_{3}\right)-\frac{1}{2}\left(\mu_{1}-\mu_{2}\right)=0 \quad 0 \end{aligned}$	$\begin{array}{lllllll} \frac{1}{2}(\lambda_0+\lambda_1-\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2) = & 1 & 2 & 2 & 1 & 1 & 0 \\ \frac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2) = & 0 & 1 & 1 & 1 & 1 & 0 \\ \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2) = & 0 & 1 & 2 & 1 & 1 & 0 \\ \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2) = & 1 & 2 & 3 & 2 & 2 & 1 \\ \frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2) = & 0 & 0 & 1 & 1 & 1 & 0. \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_ EQ1032 **[conf 0.022]**	173	0.022	\begin{aligned} & \frac{1}{2}\left(\lambda_{0}+ \lambda_{1}+\lambda_{2}-\lambda_{3}\right)+\frac{1}{2} \mu_{2}-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 2 \end{aligned} \quad 1 \quad =\frac{1}{2}\left(\lambda_{0}+\lambda_{1}-\lambda_{2}+\lambda_{3}\right)+ \frac{1}{2} \mu_{2}-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 1 \quad 1 .	*(not rendered)*	$\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2-\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 2 \quad 1 \quad 1$ $\frac{1}{2}(\lambda_0+\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 1 \quad 1$ $\frac{1}{2}(\lambda_0-\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 0 \quad 1$ $\frac{1}{2}(\lambda_0-\lambda_1-\lambda_2-\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0$ $\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2-\lambda_3)-\frac{1}{2}\mu_2+\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 2 \quad 1 \quad 2$ $\frac{1}{2}(\lambda_0+\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}\mu_2+\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 1 \quad 1 \quad 2$ $\frac{1}{2}(\lambda_0-\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 1 \quad 0 \quad 2$ $\frac{1}{2}(\lambda_0-\lambda_1-\lambda_2-\lambda_3)-\frac{1}{2}\mu_2+\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1$ $\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 3 \quad 3 \quad 2 \quad 1 \quad 1$ $\frac{1}{2}(\lambda_0+\lambda_1-\lambda_2-\lambda_3)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0$ $\frac{1}{2}(\lambda_0-\lambda_1+\lambda_2-\lambda_3)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0$ $\frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0$ $\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}\mu_3+\frac{2}{3}\nu=1 \quad 3 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 3$ $\frac{1}{2}(\lambda_0+\lambda_1-\lambda_2-\lambda_3)-\frac{1}{2}\mu_3+\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 1 \quad 1 \quad 2$ $\frac{1}{2}(\lambda_0-\lambda_1+\lambda_2-\lambda_3)-\frac{1}{2}\mu_3+\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 1 \quad 0 \quad 2$ $\frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}\mu_3+\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 0 \quad 0 \quad 2$

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_ EQ1033 **[conf 0.026]**	174	■ 0.026	`\begin{array}{lllllllll} \lambda_{0}+\frac{1}{2} \mu_{1}+ \frac{1}{3} \nu+r=2 & 4 & 6 & 4 & 3 & 2 & 1 & 3 & \\ \lambda_{0}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu-r=2 & 4 & 5 & 4 & 3 & 2 & 1 & 3 \\ \lambda_{0}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu+r=2 & 3 & 5 & 4 & 3 & 2 & 1 & 2 \\ \lambda_{0}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu-r=2 & 3 & 4 & 4 & 3 & 2 & 1 & 2 \\ \lambda_{1}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu+r=0 & 1 & 2 & 1 & 1 & 1 & 1 & 1 \\ \lambda_{1}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu-r=0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_{1}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu+r=0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ \lambda_{1}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu-r=0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ \lambda_{2}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu+r=0 & 1 & 2 & 1 & 1 & 1 & 0 & 1 \\ \lambda_{2}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu-r=0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ \lambda_{2}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu+r=0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_{2}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu-r=0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & \\ \lambda_{3}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu+r=0 & 1 & 2 & 1 & 1 & 0 & 0 & 1 & \\ \lambda_{3}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu-r=0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & \\ \lambda_{3}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu+r=0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & \\ \lambda_{3}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu-r=0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & \\ 1+\lambda_{2}-\lambda_{3}+ \frac{1}{2} \mu_{2}+\frac{1}{3} \nu+r=1 & 2 & 4 & 3 & 2 & 2 & 1 \\ 1+\lambda_{2}-\lambda_{3}+\frac{1}{2} \mu_{2}+\frac{1}{3} \nu-r=1 & 2 & 3 & 3 & 2 & 2 & 1 & 2 \end{array}`	$\begin{array}{lllllllll} \lambda_{0}+\frac{1}{2} \mu_{1}+ \frac{1}{3} \nu+r=2 & 4 & 6 & 4 & 3 & 2 & 1 & 3 \\ \lambda_{0}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu-r=2 & 4 & 5 & 4 & 3 & 2 & 1 & 3 \\ \lambda_{0}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu+r=2 & 3 & 5 & 4 & 3 & 2 & 1 & 2 \\ \lambda_{0}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu-r=2 & 3 & 4 & 4 & 3 & 2 & 1 & 2 \\ \lambda_{1}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu+r=0 & 1 & 2 & 1 & 1 & 1 & 1 & 1 \\ \lambda_{1}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu-r=0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_{1}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu+r=0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ \lambda_{1}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu-r=0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ \lambda_{2}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu+r=0 & 1 & 2 & 1 & 1 & 1 & 0 & 1 \\ \lambda_{2}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu-r=0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ \lambda_{2}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu+r=0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_{2}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu-r=0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ \lambda_{3}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu+r=0 & 1 & 2 & 1 & 1 & 0 & 0 & 1 \\ \lambda_{3}+\frac{1}{2} \mu_{1}+\frac{1}{3} \nu-r=0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ \lambda_{3}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu+r=0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_{3}-\frac{1}{2} \mu_{1}-\frac{1}{3} \nu-r=0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1+\lambda_{2}-\lambda_{3}+ \frac{1}{2} \mu_{2}+\frac{1}{3} \nu+r=1 & 2 & 4 & 3 & 2 & 2 & 1 \\ 1+\lambda_{2}-\lambda_{3}+\frac{1}{2} \mu_{2}+\frac{1}{3} \nu-r=1 & 2 & 3 & 3 & 2 & 2 & 1 & 2 \end{array}$	$\begin{array}{llllllll} \lambda_0+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=2 & 4 & 6 & 4 & 3 & 2 & 1 & 3 \\ \lambda_0+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=2 & 4 & 5 & 4 & 3 & 2 & 1 & 3 \\ \lambda_0-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=2 & 3 & 5 & 4 & 3 & 2 & 1 & 2 \\ \lambda_0-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=2 & 3 & 4 & 4 & 3 & 2 & 1 & 2 \\ \lambda_1+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=0 & 1 & 2 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ \lambda_1-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ \lambda_2+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=0 & 1 & 2 & 1 & 1 & 1 & 0 & 1 \\ \lambda_2+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ \lambda_2-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_2-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ \lambda_3+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=0 & 1 & 2 & 1 & 1 & 0 & 0 & 1 \\ \lambda_3+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ \lambda_3-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_3-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ {}_1+\lambda_2-\lambda_3)+\frac{1}{2}\mu_2+\frac{1}{3}\nu+r=1 & 2 & 4 & 3 & 2 & 2 & 1 & 2 \\ {}_1+\lambda_2-\lambda_3)+\frac{1}{2}\mu_2+\frac{1}{3}\nu-r=1 & 2 & 3 & 3 & 2 & 2 & 1 & 2 \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_ EQ1031 **[conf 0.036]**	173	■ 0.036	`\begin{aligned} & \lambda_{2}+\frac{1}{2} \mu_{1}-\frac{2}{3} \nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_{2}-\frac{1}{2} \mu_{1}+\frac{2}{3} \nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_{3}+\frac{1}{2} \mu_{1}-\frac{2}{3} \nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \\ & \lambda_{3}-\frac{1}{2} \mu_{1}+\frac{2}{3} \nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ & 0 \end{aligned}`	$\lambda_{2}+\frac{1}{2} \mu_{1}-\frac{2}{3} \nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0$ $\lambda_{2}-\frac{1}{2} \mu_{1}+\frac{2}{3} \nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0$ $\lambda_{3}+\frac{1}{2} \mu_{1}-\frac{2}{3} \nu=0 \quad 1 \quad 1 \quad 1 \quad 1$ $\lambda_{3}-\frac{1}{2} \mu_{1}+\frac{2}{3} \nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0$ 0	$\lambda_{2}+\frac{1}{2} \mu_{1}-\frac{2}{3} \nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0$ $\lambda_{2}-\frac{1}{2} \mu_{1}+\frac{2}{3} \nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1$ $\lambda_{3}+\frac{1}{2} \mu_{1}-\frac{2}{3} \nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0$ $\lambda_{3}-\frac{1}{2} \mu_{1}+\frac{2}{3} \nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1$
0902.0431_ EQ0761 **[conf 0.036]**	127	■ 0.036	`\begin{aligned} & \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}+\lambda_{2}-\lambda_{3}\right)+\frac{1}{2} \mu_{2}-\frac{2}{3} \nu=0 \quad 0 \quad 1 \quad 1 \quad` **[U+0930] [U+094D]** `0 \quad 0 \\ & \frac{1}{2}\left(\lambda_{0}+\lambda_{1}+\lambda_{2}-\lambda_{3}\right)+\frac{1}{2} \mu_{2}-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \end{aligned} \quad 1 \quad 2 .`	$\frac{1}{2}\left(-\lambda_{0}-\lambda_{1}+\lambda_{2}-\lambda_{3}\right)+\frac{1}{2} \mu_{2}-\frac{2}{3} \nu=0 \quad 0 \quad 1 \quad 1$ **[U+0930][U+094D]** $0 \quad 0$ $\frac{1}{2}\left(\lambda_{0}+\lambda_{1}+\lambda_{2}-\lambda_{3}\right)+\frac{1}{2} \mu_{2}-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 2$ $\quad 1 \quad 2.$	$\frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2-\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1$ $\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2-\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 2$ $\frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=0 \quad 1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 2$ $\frac{1}{2}(\lambda_0-\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 2$ $\frac{1}{2}(\lambda_0+\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 2 \quad 2 \quad 3 \quad 2 \quad 1 \quad 2$ $\frac{1}{2}(-\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1$ $\frac{1}{2}(\lambda_0-\lambda_1-\lambda_2-\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1$ $\frac{1}{2}(-\lambda_0+\lambda_1-\lambda_2-\lambda_3)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1$ $\frac{1}{2}(\lambda_0-\lambda_1+\lambda_2-\lambda_3)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 0 \quad 1$ $\frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_3+\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 2 \quad 1 \quad 0 \quad 1$ $\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}\mu_3-\frac{2}{3}\nu=0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0$ $\frac{1}{2}(\lambda_0+\lambda_1-\lambda_2-\lambda_3)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 0 \quad 1$ $\frac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=0 \quad 1 \quad 1 \quad 2 \quad 1 \quad 0 \quad 1$ $\frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=0 \quad 1 \quad 2 \quad 2 \quad 1 \quad 0 \quad 1$ $\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 4 \quad 3 \quad 1 \quad 2$ $\frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=0 \quad 0 \quad 1 \quad 2 \quad 1 \quad 0 \quad 1.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_ EQ0516 **[conf 0.041]**	085	■ 0.041	`\begin{aligned} & \lambda_{0}+\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=1 \quad 1 \quad 1 \quad 1 \quad a \quad 0 \\ & \lambda_{1}+\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 1 \quad 1 \quad 1 \quad a \\ & \lambda_{2}+\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 0 \quad 1 \quad 1 \\ & \lambda_{3}+\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 0 \quad 0 \\ & \lambda_{0}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_{1}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_{2}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 0 \quad 1 \quad 1 \\ & \lambda_{3}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 0 \quad 0 \quad 0 \end{aligned}`	$\lambda_0+\frac{1}{2}(\mu_2-\mu_3)=1 \quad 1 \quad 1 \quad 1 \quad a \quad 0$ $\lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 \quad 1 \quad 1 \quad 1 \quad a$ $\lambda_2+\frac{1}{2}(\mu_2-\mu_3)=0 \quad 0 \quad 1 \quad 1$ $\lambda_3+\frac{1}{2}(\mu_2-\mu_3)=0 \quad 0 \quad 0$ $\lambda_0-\frac{1}{2}(\mu_2-\mu_3)=1 \quad 1 \quad 1 \quad 1 \quad 0$ $\lambda_1-\frac{1}{2}(\mu_2-\mu_3)=0 \quad 1 \quad 1 \quad 1 \quad 0$ $\lambda_2-\frac{1}{2}(\mu_2-\mu_3)=0 \quad 0 \quad 1 \quad 1$ $\lambda_3-\frac{1}{2}(\mu_2-\mu_3)=0 \quad 0 \quad 0 \quad 0$	$\lambda_0+\frac{1}{2}(\mu_2-\mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0$ $\lambda_1+\frac{1}{2}(\mu_2-\mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0$ $\lambda_2+\frac{1}{2}(\mu_2-\mu_3) = 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0$ $\lambda_3+\frac{1}{2}(\mu_2-\mu_3) = 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0$ $\lambda_0-\frac{1}{2}(\mu_2-\mu_3) = 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1$ $\lambda_1-\frac{1}{2}(\mu_2-\mu_3) = 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1$ $\lambda_2-\frac{1}{2}(\mu_2-\mu_3) = 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1$ $\lambda_3-\frac{1}{2}(\mu_2-\mu_3) = 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1$
0902.0431_ EQ0425 **[conf 0.082]**	070	■ 0.082	`\begin{aligned} & A X A^{*} \times A Y A^{*}=A(X \times Y) A^{*} \\ & \left(P M A^{*}\right)^{*}\left(P N A^{*}\right)=A M^{*} P^{*} P N A=A\left(M^{*} N\right) A, \\ & \left(P M A^{*}\right)\left(A Y A^{*}\right)=P(M Y) A^{*}, \\ & \overline{P M A^{*} \times P N A^{*}}= \frac{\bar{t} \widetilde{P}(M \times N)^{t} \widetilde{A} *}{}=P \overline{M \times N} A^{*}, \text { etc. } \end{aligned}`	$AXA^{*}\times AYA^{*}=A(X\times Y)A^{*}$ $(PMA^{*})^{*}(PNA^{*})=AM^{*}P^{*}PNA=A(M^{*}N)A,$ $(PMA^{*})(AYA^{*})=P(MY)A^{*},$ $\overline{PMA^{*}\times PNA^{*}}=\frac{\bar{t}\widetilde{P}(M\times N)^{t}\widetilde{A}*}{}=P\overline{M\times N}A^{*}, \text{ etc.}$	$AXA^{*}\times AYA^{*}=A(X\times Y)A^{*},$ $(PMA^{*})^{*}(PNA^{*})=AM^{*}P^{*}PNA=A(M^{*}N)A,$ $(PMA^{*})(AYA^{*})=P(MY)A^{*},$ $\overline{PMA^{*}\times PNA^{*}}=\overline{{}^{t}\widetilde{P}(M\times N){}^{t}\widetilde{A}^{*}}=P\overline{M\times N}A^{*}, \text{ etc.}$
0902.0431_ EQ1059 **[conf 0.093]**	180	■ 0.093	`\begin{aligned} & =\left\{\alpha \in\left(E_{8}{ }^{C}\right)_{1,1-, 1-} \mid \tau \widetilde{\lambda} \alpha=\alpha \tau \widetilde{\lambda}\right\} \\ & =\left\{\alpha \in E_{7}{ }^{C} \mid \tau \widetilde{\lambda} \alpha=\alpha \tau \widetilde{\lambda}\right\} \text { (Proposition 5.7.1) } \\ & =\left\{\alpha \in E_{7}{ }^{C} \mid \tau \lambda \alpha=\alpha \tau \lambda\right\} \text { (by the correspondence to Proposition 5.7.1) } \\ & =E_{7} \text { (Lemma 4.3.3.(4)). } \end{aligned}`	$=\left\{\alpha\in(E_8{}^C)_{1,1-,1-}\mid\tau\widetilde{\lambda}\alpha=\alpha\tau\widetilde{\lambda}\right\}$ $=\left\{\alpha\in E_7{}^C\mid\tau\widetilde{\lambda}\alpha=\alpha\tau\widetilde{\lambda}\right\}$ (Proposition 5.7.1) $=\left\{\alpha\in E_7{}^C\mid\tau\lambda\alpha=\alpha\tau\lambda\right\}$ (by the correspondence to Proposition 5.7.1) $=E_7$ (Lemma 4.3.3.(4)).	$=\{\alpha\in(E_8{}^C)_{1,1^-,1_-}\mid\tau\widetilde{\lambda}\alpha=\alpha\tau\widetilde{\lambda}\}$ $=\{\alpha\in E_7{}^C\mid\tau\widetilde{\lambda}\alpha=\alpha\tau\widetilde{\lambda}\}$ (Proposition 5.7.1) $=\{\alpha\in E_7{}^C\mid\tau\lambda\alpha=\alpha\tau\lambda\}$ (by the correspondence to Proposition 5.7.1) $=E_7$ (Lemma 4.3.3.(4)).

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_ EQ0757 **[conf 0.095]**	126	■ 0.095	`\begin{aligned} & \lambda_{0}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_{1}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad a \\ & \lambda_{2}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 0 \quad 1 \quad 1 \quad 1 \\ & \lambda_{3}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 0 \quad 0 \\ & 0 \end{aligned}`	$\begin{aligned} & \lambda_{0}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_{1}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad a \\ & \lambda_{2}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 0 \quad 1 \quad 1 \quad 1 \\ & \lambda_{3}-\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 \quad 0 \quad 0 \\ & 0 \end{aligned}$	$\begin{aligned} & \lambda_{0}-\frac{1}{2}(\mu_{2}-\mu_{3})=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_{1}-\frac{1}{2}(\mu_{2}-\mu_{3})=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_{2}-\frac{1}{2}(\mu_{2}-\mu_{3})=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_{3}-\frac{1}{2}(\mu_{2}-\mu_{3})=0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \end{aligned}$
0902.0431_EQ0922	153	■ 0.102	`f^{-1}\left(\sum_{i<j<k} x_{i j k} \boldsymbol{e}_{i} \wedge \boldsymbol{e}_{j} \wedge \boldsymbol{e}_{k} \right)=\left(\begin{array}{c} \left(\begin{array}{ccc} h\left(x_{156}\right) & h\left(x_{164}, \bar{x}_{256} \right) & h\left(x_{145}, \bar{x}_{356}\right) \\ h \left(x_{256}, \bar{x}_{164}\right) & h\left(x_{264} \right) & h\left(x_{245}, \bar{x}_{364}\right) \\ h \left(x_{356}, \bar{x}_{145}\right) & h\left(x_{364}, \bar{x}_{245}\right) & h\left(x_{345}\right) \end{array} \right) \\ \left(\begin{array}{ccc} h\left(x_{423}\right) & h\left(x_{431}, \bar{x}_{523}\right) & h\left(x_{412}, \bar{x}_{623}\right) \\ h\left(x_{523}, \bar{x}_{431} \right) & h\left(x_{531}\right) & h\left(x_{512}, \bar{x}_{631}\right) \\ h\left(x_{623}, \bar{x}_{412} \right) & h\left(x_{631}, \bar{x}_{512}\right) & h \left(x_{612}\right) \end{array}\right) \\ \\ \\ \left(\begin{array}{cc} h\left(x_{123}\right) & \\ h \left(x_{456}\right) \end{array}\right. \end{array} \right),`	*(not rendered)*	$f^{-1}\Big(\sum_{i<j<k} x_{ijk}\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\Big) = \left(\begin{array}{c} \left(\begin{array}{ccc} h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \end{array}\right) \\ \left(\begin{array}{ccc} h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612}) \end{array}\right) \\ h(x_{123}) \\ h(x_{456}) \end{array}\right),$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ1117	189	■ 0.108	`\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y})= \left(X_{1}, Y_{1}\right)+\left(X_{2}, Y_{2}\right)+ \left(X_{3}, Y_{3}\right) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y}\rangle=\left\langle X_{1}, Y_{1}\right\rangle+\left\langle X_{2}, Y_{2} \right\rangle+\left\langle X_{3}, Y_{3}\right\rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y}=\left(\begin{array}{ll} X_{2} \times Y_{3}-Y_{2} \times X_{3} \\ X_{3} \times Y_{1}-Y_{3} \times X_{1} \\ X_{1} \times Y_{2}-Y_{1} \times X_{2} \end{array}\right) \in\left(\mathfrak{J}^{C}\right)^{3}, \\ & \boldsymbol{X} \cdot \boldsymbol{Y}=\left(\begin{array}{ll} \left(X_{1}, Y_{1} \right) & \left(X_{1}, Y_{2}\right) \\ \left(X_{2}, Y_{1} \right) & \left(X_{2}, Y_{2}\right) \\ \left(X_{3}, Y_{1}\right) & \left(X_{2}, Y_{3}\right) \\ X \vee \boldsymbol{Y} & \left(X_{3}, Y_{3}\right) \end{array}\right)-\frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y}) E \in \mathfrak{s} \mathfrak{l}(3, C), \\ & \boldsymbol{X} Y_{1}+X_{2} \vee Y_{2}+X_{3} \vee Y_{3} \in \mathfrak{e}_{6}{ }^{C}, \end{aligned}`	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y})= \left(X_{1}, Y_{1}\right)+\left(X_{2}, Y_{2}\right)+ \left(X_{3}, Y_{3}\right) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y}\rangle=\left\langle X_{1}, Y_{1}\right\rangle+\left\langle X_{2}, Y_{2} \right\rangle+\left\langle X_{3}, Y_{3}\right\rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y}=\left(\begin{array}{ll} X_{2} \times Y_{3}-Y_{2} \times X_{3} \\ X_{3} \times Y_{1}-Y_{3} \times X_{1} \\ X_{1} \times Y_{2}-Y_{1} \times X_{2} \end{array}\right) \in\left(\mathfrak{J}^{C}\right)^{3}, \\ & \boldsymbol{X} \cdot \boldsymbol{Y}=\left(\begin{array}{ll} \left(X_{1}, Y_{1} \right) & \left(X_{1}, Y_{2}\right) \\ \left(X_{2}, Y_{1} \right) & \left(X_{2}, Y_{2}\right) \\ \left(X_{3}, Y_{1}\right) & \left(X_{2}, Y_{3}\right) \\ X \vee \boldsymbol{Y} & \left(X_{3}, Y_{3}\right) \end{array}\right)-\frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y}) E \in \mathfrak{s} \mathfrak{l}(3, C), \\ & \boldsymbol{X} Y_{1}+X_{2} \vee Y_{2}+X_{3} \vee Y_{3} \in \mathfrak{e}_{6}{ }^{C}, \end{aligned}$	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y}) E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6{}^C, \end{aligned}$

Conf. MathPix confidence: ■ $\geq$0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ0756	125	0.119	\begin{array}{lllllllllllll} \lambda_{0}-\lambda_{1}=1 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{0}+\lambda_{1}=1 & 2 & 2 & 2 & 2 & 1 \\ \lambda_{0}-\lambda_{2}=1 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_{0}+\lambda_{2}=1 & 1 & 2 & 2 & 2 & 1 \\ \lambda_{0}-\lambda_{3}=1 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_{0}+\lambda_{3}=1 & 1 & 1 & 2 & 2 & 1 \\ \lambda_{1}-\lambda_{2}=0 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_{1}+\lambda_{2}=0 & 1 & 2 & 2 & 2 & 1 \\ \lambda_{1}-\lambda_{3}=0 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_{1}+\lambda_{3}=0 & 1 & 1 & 2 & 2 & 1 \\ \lambda_{2}-\lambda_{3}=0 & 0 & 1 & 0 & 0 & 0 & 0 & \lambda_{2}+\lambda_{3}=0 & 0 & 1 & 2 & 2 & 1 \\ & & \lambda_{0}+\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=1 & 1 & 1 & 1 & 1 & 1 & & & \\ & & \lambda_{1}+\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 & 1 & 1 & 1 & 1 & 1 & & & \\ & & \lambda_{2}+\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 & 0 & 1 & 1 & 1 & 1 & & & \\ & & + \frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 & 0 & 0 & 1 & 1 & 1 & & & \end{array}	$\begin{array}{lllllllllllll} \lambda_{0}-\lambda_{1}=1 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{0}+\lambda_{1}=1 & 2 & 2 & 2 & 2 & 1 \\ \lambda_{0}-\lambda_{2}=1 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_{0}+\lambda_{2}=1 & 1 & 2 & 2 & 2 & 1 \\ \lambda_{0}-\lambda_{3}=1 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_{0}+\lambda_{3}=1 & 1 & 1 & 2 & 2 & 1 \\ \lambda_{1}-\lambda_{2}=0 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_{1}+\lambda_{2}=0 & 1 & 2 & 2 & 2 & 1 \\ \lambda_{1}-\lambda_{3}=0 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_{1}+\lambda_{3}=0 & 1 & 1 & 2 & 2 & 1 \\ \lambda_{2}-\lambda_{3}=0 & 0 & 1 & 0 & 0 & 0 & 0 & \lambda_{2}+\lambda_{3}=0 & 0 & 1 & 2 & 2 & 1 \\ & & \lambda_{0}+\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=1 & 1 & 1 & 1 & 1 & 1 & & & \\ & & \lambda_{1}+\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 & 1 & 1 & 1 & 1 & 1 & & & \\ & & \lambda_{2}+\frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 & 0 & 1 & 1 & 1 & 1 & & & \\ & & + \frac{1}{2}\left(\mu_{2}-\mu_{3}\right)=0 & 0 & 0 & 1 & 1 & 1 & & & \end{array}$	$\begin{array}{llllllllllllll} \lambda_0-\lambda_1=1 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_0+\lambda_1=1 & 2 & 2 & 2 & 2 & 1 & 1 \\ \lambda_0-\lambda_2=1 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_0+\lambda_2=1 & 1 & 2 & 2 & 2 & 1 & 1 \\ \lambda_0-\lambda_3=1 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_0+\lambda_3=1 & 1 & 1 & 2 & 2 & 1 & 1 \\ \lambda_1-\lambda_2=0 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_1+\lambda_2=0 & 1 & 2 & 2 & 2 & 1 & 1 \\ \lambda_1-\lambda_3=0 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_1+\lambda_3=0 & 1 & 1 & 2 & 2 & 1 & 1 \\ \lambda_2-\lambda_3=0 & 0 & 1 & 0 & 0 & 0 & 0 & \lambda_2+\lambda_3=0 & 0 & 1 & 2 & 2 & 1 & 1 \end{array}$ $\begin{array}{lllllll} \lambda_0+\frac{1}{2}(\mu_2-\mu_3)=1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 1 & 1 & 1 & 1 & 1 \\ \lambda_3+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 0 & 1 & 1 & 1 & 1 \end{array}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ1187	200	■ 0.183	`\begin{aligned} &=(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{a}, *(\boldsymbol{b} \wedge \boldsymbol{x}))- \left(\boldsymbol{a}_{1}, \boldsymbol{w}\right) \left(\boldsymbol{w} \wedge \boldsymbol{a}_{2}, *(\boldsymbol{b} \wedge \boldsymbol{x})\right) \\ &+ \left(\boldsymbol{a}_{2}, \boldsymbol{w}\right) \left(\boldsymbol{w} \wedge \boldsymbol{a}_{1}, *(\boldsymbol{b} \wedge \boldsymbol{x})\right)- \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \\ &= \frac{2}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w})- \left(\boldsymbol{a}_{1}, \boldsymbol{w}\right) \left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_{2}, \boldsymbol{e}_{1} \wedge \cdots \wedge \boldsymbol{e}_{5}\right) \\ &+\left(\boldsymbol{a}_{2}, \boldsymbol{w}\right)\left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_{1}, \boldsymbol{e}_{1} \wedge \cdots \wedge \boldsymbol{e}_{5}\right) \\ &((\boldsymbol{b} \times □ □ 一 □ 日 一 \zh20 \\ &=(\boldsymbol{a} \wedge \boldsymbol{x})) \boldsymbol{v}, \boldsymbol{w}) \\ &=(\boldsymbol{v}, \boldsymbol{w})(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{v})+\left(\boldsymbol{a}_{1}, \boldsymbol{w} \right)\left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_{2}, \boldsymbol{e}_{1} \wedge \cdots \wedge \boldsymbol{e}_{5}\right) \\ &-\left(\boldsymbol{a}_{2}, \boldsymbol{w}\right)\left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_{1}, \boldsymbol{e}_{1} \wedge \cdots \wedge \boldsymbol{e}_{5}\right)-\frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) . \end{aligned}`	*(not rendered)*	$\begin{aligned} &= (\boldsymbol{v},\boldsymbol{w})(\boldsymbol{a},*(\boldsymbol{b}\wedge\boldsymbol{x})) - (\boldsymbol{a}_1,\boldsymbol{w})(\boldsymbol{w}\wedge\boldsymbol{a}_2,*(\boldsymbol{b}\wedge\boldsymbol{x})) \\ &\quad +(\boldsymbol{a}_2,\boldsymbol{w})(\boldsymbol{w}\wedge\boldsymbol{a}_1,*(\boldsymbol{b}\wedge\boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a}\wedge\boldsymbol{b}),\boldsymbol{x})(\boldsymbol{v},\boldsymbol{w}) \\ &= \frac{2}{5}(*(\boldsymbol{a}\wedge\boldsymbol{b}),\boldsymbol{x})(\boldsymbol{v},\boldsymbol{w}) - (\boldsymbol{a}_1,\boldsymbol{w})(\boldsymbol{b}\wedge\boldsymbol{x}\wedge\boldsymbol{w}\wedge\boldsymbol{a}_2,\boldsymbol{e}_1\wedge\cdots\wedge\boldsymbol{e}_5) \\ &\quad +(\boldsymbol{a}_2,\boldsymbol{w})(\boldsymbol{b}\wedge\boldsymbol{x}\wedge\boldsymbol{w}\wedge\boldsymbol{a}_1,\boldsymbol{e}_1\wedge\cdots\wedge\boldsymbol{e}_5), \\ &((\boldsymbol{b}\times *(\boldsymbol{a}\wedge\boldsymbol{x}))\boldsymbol{v},\boldsymbol{w}) \\ &= (\boldsymbol{x}\wedge\boldsymbol{a},\boldsymbol{w}\wedge *(\boldsymbol{b}\wedge\boldsymbol{v})) - \frac{3}{5}(*(\boldsymbol{a}\wedge\boldsymbol{b}),\boldsymbol{x})(\boldsymbol{v},\boldsymbol{w}) \\ &= (\boldsymbol{v},\boldsymbol{w})(*(\boldsymbol{a}\wedge\boldsymbol{b}),\boldsymbol{v}) + (\boldsymbol{a}_1,\boldsymbol{w})(\boldsymbol{b}\wedge\boldsymbol{x}\wedge\boldsymbol{w}\wedge\boldsymbol{a}_2,\boldsymbol{e}_1\wedge\cdots\wedge\boldsymbol{e}_5) \\ &\quad -(\boldsymbol{a}_2,\boldsymbol{w})(\boldsymbol{b}\wedge\boldsymbol{x}\wedge\boldsymbol{w}\wedge\boldsymbol{a}_1,\boldsymbol{e}_1\wedge\cdots\wedge\boldsymbol{e}_5) - \frac{3}{5}(*(\boldsymbol{a}\wedge\boldsymbol{b}),\boldsymbol{x})(\boldsymbol{v},\boldsymbol{w}). \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ0830	138	■ 0.186	`\begin{aligned} \kappa(X, Y, \xi, \eta)= & \kappa \left(\left(\begin{array}{ccc} \xi_{1} & x_{3} & \bar{x}_{2} \\ \bar{x}_{3} & \xi_{2} & x_{1} \\ x_{2} & \bar{x}_{1} & \xi_{3} \end{array}\right),\left(\begin{array}{ccc} \eta_{1} & y_{3} & \bar{y}_{2} \\ \bar{y}_{3} & \eta_{2} & y_{1} \\ y_{2} & \bar{y}_{1} & \eta_{3} \end{array}\right), \xi, \eta\right) \\ = & \left(\left(\begin{array}{ccc} -\xi_{1} & 0 & 0 \\ 0 & \xi_{2} & x_{1} \\ 0 & \bar{x}_{1} & \xi_{3} \end{array} \right),\left(\begin{array}{ccc} \eta_{1} & 0 & 0 \\ 0 & -\eta_{2} & -y_{1} \\ 0 & -\bar{y}_{1} & -\eta_{3} \end{array}\right),-\xi, \eta\right), \\ \mu(X, Y, \xi, \eta)= & \left(\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & \eta_{3} & -y_{1} \\ 0 & -\bar{y}_{1} & \eta_{2} \end{array}\right),\left(\begin{array}{ccc} \xi & 0 & 0 \\ 0 & \xi_{3} & -x_{1} \\ 0 & -\bar{x}_{1} & \xi_{2} \end{array}\right), \eta_{1}, \xi_{1}\right) . \end{aligned}`	$\begin{aligned} \kappa(X, Y, \xi, \eta)= & \kappa \left(\left(\begin{array}{ccc} \xi_{1} & x_{3} & \bar{x}_{2} \\ \bar{x}_{3} & \xi_{2} & x_{1} \\ x_{2} & \bar{x}_{1} & \xi_{3} \end{array}\right),\left(\begin{array}{ccc} \eta_{1} & y_{3} & \bar{y}_{2} \\ \bar{y}_{3} & \eta_{2} & y_{1} \\ y_{2} & \bar{y}_{1} & \eta_{3} \end{array}\right), \xi, \eta\right) \\ = & \left(\left(\begin{array}{ccc} -\xi_{1} & 0 & 0 \\ 0 & \xi_{2} & x_{1} \\ 0 & \bar{x}_{1} & \xi_{3} \end{array} \right),\left(\begin{array}{ccc} \eta_{1} & 0 & 0 \\ 0 & -\eta_{2} & -y_{1} \\ 0 & -\bar{y}_{1} & -\eta_{3} \end{array}\right),-\xi, \eta\right), \\ \mu(X, Y, \xi, \eta)= & \left(\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & \eta_{3} & -y_{1} \\ 0 & -\bar{y}_{1} & \eta_{2} \end{array}\right),\left(\begin{array}{ccc} \xi & 0 & 0 \\ 0 & \xi_{3} & -x_{1} \\ 0 & -\bar{x}_{1} & \xi_{2} \end{array}\right), \eta_{1}, \xi_{1}\right) . \end{aligned}$	$\begin{aligned} \kappa(X, Y, \xi, \eta) &= \kappa\Big(\begin{pmatrix} \xi_1 & x_3 & \overline{x}_2 \\ \overline{x}_3 & \xi_2 & x_1 \\ x_2 & \overline{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & y_3 & \overline{y}_2 \\ \overline{y}_3 & \eta_2 & y_1 \\ y_2 & \overline{y}_1 & \eta_3 \end{pmatrix}, \xi, \eta\Big) \\ &= \Big(\begin{pmatrix} -\xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \overline{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & 0 & 0 \\ 0 & -\eta_2 & -y_1 \\ 0 & -\overline{y}_1 & -\eta_3 \end{pmatrix}, -\xi, \eta\Big), \\ \mu(X, Y, \xi, \eta) &= \Big(\begin{pmatrix} \eta & 0 & 0 \\ 0 & \eta_3 & -y_1 \\ 0 & -\overline{y}_1 & \eta_2 \end{pmatrix}, \begin{pmatrix} \xi & 0 & 0 \\ 0 & \xi_3 & -x_1 \\ 0 & -\overline{x}_1 & \xi_2 \end{pmatrix}, \eta_1, \xi_1\Big). \end{aligned}$
0902.0431_EQ1041	177	■ 0.204	`\begin{aligned} & \left(\alpha_{i}, \alpha_{i}\right)= \frac{1}{30}, \quad i=1,2,3,4,5,6,7,8 \\ & \left(\alpha_{i}, \alpha_{i+1}\right)=-\frac{1}{60}, \quad i=1,2, \cdots, 6, \quad\left(\alpha_{3}, \alpha_{8} \right)=-\frac{1}{60} \\ & \left(\alpha_{i}, \alpha_{j} \right)=0, \quad \text { otherwise } \\ & (-\mu,-\mu)= \frac{1}{30}, \quad\left(-\mu, \alpha_{7}\right)=- \frac{1}{60}, \quad\left(-\mu, \alpha_{i}\right)=0, \quad i=1,2,3,4,5,6,8, \end{aligned}`	$\begin{aligned} & \left(\alpha_{i}, \alpha_{i}\right)= \frac{1}{30}, \quad i=1,2,3,4,5,6,7,8 \\ & \left(\alpha_{i}, \alpha_{i+1}\right)=-\frac{1}{60}, \quad i=1,2, \cdots, 6, \quad\left(\alpha_{3}, \alpha_{8} \right)=-\frac{1}{60} \\ & \left(\alpha_{i}, \alpha_{j} \right)=0, \quad \text { otherwise } \\ & (-\mu,-\mu)= \frac{1}{30}, \quad\left(-\mu, \alpha_{7}\right)=- \frac{1}{60}, \quad\left(-\mu, \alpha_{i}\right)=0, \quad i=1,2,3,4,5,6,8, \end{aligned}$	$\begin{aligned} &(\alpha_i, \alpha_i) = \frac{1}{30}, \quad i = 1,2,3,4,5,6,7,8, \\ &(\alpha_i, \alpha_{i+1}) = -\frac{1}{60}, \quad i = 1,2,\cdots,6, \quad (\alpha_3, \alpha_8) = -\frac{1}{60}, \\ &(\alpha_i, \alpha_j) = 0, \quad \text{otherwise}, \\ &(-\mu, -\mu) = \frac{1}{30}, \quad (-\mu, \alpha_7) = -\frac{1}{60}, \quad (-\mu, \alpha_i) = 0, \quad i = 1,2,3,4,5,6,8, \end{aligned}$
0902.0431_EQ0340	054	■ 0.226	`\begin{array}{lr} \lambda_{0}=\alpha_{1}+\alpha_{2}+ & \alpha_{3} \\ \lambda_{1}= & \alpha_{2}+\alpha_{3} \\ \lambda_{3}= & \alpha_{3} \\ \lambda_{3}= & \alpha_{1}+2 \alpha_{2}+3 \alpha_{3}+2 \alpha_{4}, \end{array}`	$\begin{array}{lr} \lambda_{0}=\alpha_{1}+\alpha_{2}+ & \alpha_{3} \\ \lambda_{1}= & \alpha_{2}+\alpha_{3} \\ \lambda_{3}= & \alpha_{3} \\ \lambda_{3}= & \alpha_{1}+2 \alpha_{2}+3 \alpha_{3}+2 \alpha_{4}, \end{array}$	$\begin{aligned} \lambda_0 &= \alpha_1 + \alpha_2 + \alpha_3 \\ \lambda_1 &= \alpha_2 + \alpha_3 \\ \lambda_3 &= \alpha_3 \\ \lambda_3 &= \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4, \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ0937	155	0.238	\begin{aligned} & P=(0,0,1,0) \\ & \xrightarrow{f} \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3} \\ & \stackrel{D}{\longrightarrow} D \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}+\boldsymbol{e}_{1} \wedge D \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}+ D \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2} \wedge D \boldsymbol{e}_{3} \\ & \quad+\left(b_{11}+\frac{\nu}{3} \right) \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}-\bar{l}_{11} \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}- \bar{l}_{12} \boldsymbol{e}_{5} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}-\bar{l}_{13} \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3} \\ & \quad+\left(b_{33}+\frac{\nu}{3}\right) \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}-\bar{l}_{21} \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{3} \wedge \boldsymbol{e}_{1}-\bar{l}_{21} \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2}-\bar{l}_{32} \boldsymbol{e}_{5} \wedge \boldsymbol{e}_{3} \wedge \boldsymbol{e}_{2}-\bar{l}_{33} \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{3} \\ & \quad\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right) \\ & \xrightarrow{f^{-1}}\left(\begin{array}{ccc} -h \left(\bar{l}_{11}\right) & -h\left(\bar{l}_{21}, l_{12} \right) & -h\left(\bar{l}_{31}, l_{13}\right) \\ -h \left(\bar{l}_{12}, l_{21}\right) & -h\left(\bar{l}_{22} \right) & -h\left(\bar{l}_{32}, l_{23}\right) \\ -h \left(\bar{l}_{13}, l_{31}\right) & -h\left(\bar{l}_{23}, l_{32}\right) & -h\left(\bar{l}_{33}\right) \end{array} \right)=\left(\begin{array}{c} h(\nu) \\ 0 \\ -\tau h(L) \\ -i \nu e_{1} \\ 0 \end{array}\right) h(L) \\ & =\left(\begin{array}{ccc} \phi_{\boldsymbol{C}}(B, C)+ \frac{i \nu e_{1}}{3} & -2 \tau h(L) & 0 \\ 2 h(L) & \tau \phi_{\boldsymbol{C}}(B, C) \tau-\frac{i \nu e_{1}}{3} & -\tau h(L) \\ 0 & h(L) & 0 \\ -\tau h(L) & 0 & 0 \\ 0 \\ 1 \\ 0 \end{array}\right) \\ & =\Phi\left(\phi_{ \boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_{1}	$\begin{aligned} & P=(0,0,1,0) \\ & \xrightarrow{f} \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3} \\ & \stackrel{D}{\longrightarrow} D \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}+\boldsymbol{e}_{1} \wedge D \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}+\boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2} \wedge D \boldsymbol{e}_{3} \\ & \quad+\left(b_{11}+\frac{\nu}{3}\right) \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}-\bar{l}_{11} \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}-\bar{l}_{12} \boldsymbol{e}_{5} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}-\bar{l}_{13} \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3} \\ & \quad+\left(b_{33}+\frac{\nu}{3}\right) \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3}-\bar{l}_{21} \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{3} \wedge \boldsymbol{e}_{1}-\bar{l}_{21} \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{2}-\bar{l}_{32} \boldsymbol{e}_{5} \wedge \boldsymbol{e}_{3} \wedge \boldsymbol{e}_{2}-\bar{l}_{33} \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{3} \\ & \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right) \\ & \xrightarrow{f^{-1}}\left(\begin{array}{ccc} -h\left(\bar{l}_{11}\right) & -h\left(\bar{l}_{21}, l_{12}\right) & -h\left(\bar{l}_{31}, l_{13}\right) \\ -h\left(\bar{l}_{12}, l_{21}\right) & -h\left(\bar{l}_{22}\right) & -h\left(\bar{l}_{32}, l_{23}\right) \\ -h\left(\bar{l}_{13}, l_{31}\right) & -h\left(\bar{l}_{23}, l_{32}\right) & -h\left(\bar{l}_{33}\right) \end{array}\right)=\left(\begin{array}{c} h(\nu) \\ 0 \\ -\tau h(L) \\ -i \nu e_{1} \\ 0 \end{array}\right) h(L) \\ & =\left(\begin{array}{ccc} \phi_{\boldsymbol{C}}(B, C)+\frac{i \nu e_{1}}{3} & -2 \tau h(L) & 0 \\ 2 h(L) & \tau \phi_{\boldsymbol{C}}(B, C) \tau-\frac{i \nu e_{1}}{3} & -\tau h(L) \\ 0 & h(L) & 0 \\ -\tau h(L) & 0 & 0 \\ 0 \\ 1 \\ 0 \end{array}\right) \\ & =\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_{1}\right) P. \end{aligned}$	$\begin{aligned} & P=(0,0,1,0) \\ & \xrightarrow{f} \boldsymbol{e}_1\wedge\boldsymbol{e}_2\wedge\boldsymbol{e}_3 \\ & \xrightarrow{D} D\boldsymbol{e}_1\wedge\boldsymbol{e}_2\wedge\boldsymbol{e}_3+\boldsymbol{e}_1\wedge D\boldsymbol{e}_2\wedge\boldsymbol{e}_3+\boldsymbol{e}_1\wedge\boldsymbol{e}_2\wedge D\boldsymbol{e}_3 \\ & =\Big(b_{11}+\frac{\nu}{3}\Big)\boldsymbol{e}_1\wedge\boldsymbol{e}_2\wedge\boldsymbol{e}_3-\bar{l}_{11}\boldsymbol{e}_4\wedge\boldsymbol{e}_2\wedge\boldsymbol{e}_3-\bar{l}_{12}\boldsymbol{e}_5\wedge\boldsymbol{e}_2\wedge\boldsymbol{e}_3-\bar{l}_{13}\boldsymbol{e}_6\wedge\boldsymbol{e}_2\wedge\boldsymbol{e}_3 \\ & \quad+\Big(b_{22}+\frac{\nu}{3}\Big)\boldsymbol{e}_1\wedge\boldsymbol{e}_2\wedge\boldsymbol{e}_3-\bar{l}_{21}\boldsymbol{e}_4\wedge\boldsymbol{e}_3\wedge\boldsymbol{e}_1-\bar{l}_{22}\boldsymbol{e}_1\wedge\boldsymbol{e}_5\wedge\boldsymbol{e}_3-\bar{l}_{23}\boldsymbol{e}_1\wedge\boldsymbol{e}_6\wedge\boldsymbol{e}_3 \\ & \quad+\Big(b_{33}+\frac{\nu}{3}\Big)\boldsymbol{e}_1\wedge\boldsymbol{e}_2\wedge\boldsymbol{e}_3-\bar{l}_{31}\boldsymbol{e}_4\wedge\boldsymbol{e}_1\wedge\boldsymbol{e}_2-\bar{l}_{32}\boldsymbol{e}_5\wedge\boldsymbol{e}_1\wedge\boldsymbol{e}_2-\bar{l}_{33}\boldsymbol{e}_6\wedge\boldsymbol{e}_1\wedge\boldsymbol{e}_2 \\ & \xrightarrow{f^{-1}} \left(\begin{array}{c} \begin{pmatrix}0&0&0\\0&0&0\\0&0&0\end{pmatrix} \\ \begin{pmatrix} -h(\bar{l}_{11}) & -h(\bar{l}_{21},l_{12}) & -h(\bar{l}_{31},l_{13}) \\ -h(\bar{l}_{12},l_{21}) & -h(\bar{l}_{22}) & -h(\bar{l}_{32},l_{23}) \\ -h(\bar{l}_{13},l_{31}) & -h(\bar{l}_{23},l_{32}) & -h(\bar{l}_{33})\end{pmatrix} \\ h(\nu) \\ 0 \end{array}\right) = \begin{pmatrix}0\\-\tau h(L)\\-i\nu e_1\\0\end{pmatrix} \\ & = \begin{pmatrix} \phi_{\boldsymbol{C}}(B,C)+\frac{i\nu e_1}{3} & -2\tau h(L) & 0 & h(L) \\ 2h(L) & \tau\phi_{\boldsymbol{C}}(B,C)\tau-\frac{i\nu e_1}{3} & -\tau h(L) & 0 \\ 0 & h(L) & -i\nu e_1 & 0 \\ -\tau h(L) & 0 & 0 & i\nu e_1 \end{pmatrix}\begin{pmatrix}0\\0\\1\\0\end{pmatrix} \\ & =\Phi(\phi_{\boldsymbol{C}}(B,C),h(L),-\tau h(L),-i\nu e_1)P. \end{aligned}$

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>0902.0431_EQ0834</td><td>139</td><td>■ 0.239</td><td>\[\begin{aligned} \left(\left(\mathfrak{e}_{7}\right)^{ \kappa, \mu}\right)_{\left(0, E_{1}, 0,1\right)}=\left \{\Phi \in\left(\mathfrak{e}_{7}\right)^{\kappa, \mu} \mid \Phi\left(\left(0, E_{1}, 0,1\right)\right)=0 \right\} \\ =\left\{\begin{array}{l|l} \Phi(\phi, A,-\tau A, 0) \in \mathfrak{e}_{7} & \left\lvert\, \begin{array}{l} \phi \in \mathfrak{e}_{6}, \phi E_{1}=0 \\ A \in \mathfrak{J}^{C}, 2 E_{1} \times A=\tau A \end{array}\right. \end{array}\right\} . \end{aligned} \] \end{itemize}</td><td>$$\begin{aligned} \left(\left(\mathfrak{e}_{7}\right)^{\kappa, \mu}\right)_{\left(0, E_{1}, 0,1\right)}=\left\{\Phi \in\left(\mathfrak{e}_{7}\right)^{\kappa, \mu} \mid \Phi\left(\left(0, E_{1}, 0,1\right)\right)=0 \right\} \\ =\left\{\begin{array}{l|l} \Phi(\phi, A,-\tau A, 0) \in \mathfrak{e}_{7} & \left\lvert\, \begin{array}{l} \phi \in \mathfrak{e}_{6}, \phi E_{1}=0 \\ A \in \mathfrak{J}^{C}, 2 E_{1} \times A=\tau A \end{array}\right. \end{array}\right\} . \end{aligned}$$</td><td>$$\begin{aligned} &((\mathfrak{e}_7)^{\kappa,\mu})_{(0,E_1,0,1)} = \{\Phi \in (\mathfrak{e}_7)^{\kappa,\mu} \mid \Phi((0,E_1,0,1)) = 0\} \\ &\quad = \left\{ \Phi(\phi, A, -\tau A, 0) \in \mathfrak{e}_7 \,\middle|\, \begin{array}{l} \phi \in \mathfrak{e}_6, \phi E_1 = 0, \\ A \in \mathfrak{J}^C, 2E_1 \times A = \tau A \end{array} \right\}. \end{aligned}$$</td></tr>
<tr><td>0902.0431_EQ0946</td><td>157</td><td>■ 0.244</td><td>\begin{aligned} \psi(A, & \epsilon) \psi(B, 1) P=\psi(A, \epsilon)\left(f^{-1}(B(f P))\right) \\ & \left.=f^{-1} \left(A f \overline{\left(f^{-1}(B(f P)\right.}\right) \right)=f^{-1}\left(A f\left(f^{-1}\left(\overline{ \left.\left(\mathrm{Ad} J_{3}\right) B\right)}(f \bar{P}) \right)\right)\right. \\ & =f^{-1}((A(\epsilon B))(f \bar{P}))=\psi(A(\epsilon B), \epsilon) P \end{aligned}</td><td>$$\begin{aligned} \psi(A, & \epsilon) \psi(B, 1) P=\psi(A, \epsilon)\left(f^{-1}(B(f P))\right) \\ & =f^{-1} \left(A f \overline{\left(f^{-1}(B(f P)\right)}\right)=f^{-1}\left(A f\left(f^{-1}\left(\overline{\left(\mathrm{Ad} J_{3}\right) B)}(f \bar{P}) \right)\right)\right) \\ & =f^{-1}((A(\epsilon B))(f \bar{P}))=\psi(A(\epsilon B), \epsilon) P \end{aligned}$$</td><td>$$\begin{aligned} \psi(A,\epsilon)\psi(B,1)P &= \psi(A,\epsilon)(f^{-1}(B(fP))) \\ &= f^{-1}(Af\overline{(f^{-1}(B(fP))})) = f^{-1}(Af(f^{-1}(\overline{((\mathrm{Ad}J_3)B)}(f\overline{P}))) \\ &= f^{-1}((A(\epsilon B))(f\overline{P})) = \psi(A(\epsilon B),\epsilon)P. \end{aligned}$$</td></tr>
<tr><td>0902.0431_EQ1037</td><td>176</td><td>■ 0.252</td><td>\begin{aligned} B_{8}\left(\widetilde{h}, \widetilde{h}^{ \prime}\right) & =\frac{5}{3} B_{7}\left(\Phi(h), \Phi\left(h^{\prime}\right)\right)+120 r r^{\prime}(\text { Theorem 5.3.2 }) \\ & =\frac{5}{3} 6\left(6 \sum_{k=0}^{3} \lambda_{k} \lambda_{k}{ }^{\prime}+3 \sum_{j=1}^{3} \mu_{j} \mu_{j}{ }^{\prime}+4 \nu \nu^{\prime}\right)+120 r r^{\prime} \quad(\text { Theorem 4.6.2 }) \\ & =60 \sum_{k=0}^{3} \lambda_{k} \lambda_{k}{ }^{\prime}+30 \sum_{j=1}^{3} \mu_{j} \mu_{j}{ }^{\prime}+40 \nu \nu^{\prime}+120 r r^{\prime} . \end{aligned}</td><td>$$\begin{aligned} B_{8}\left(\widetilde{h}, \widetilde{h}^{\prime}\right) & =\frac{5}{3} B_{7}\left(\Phi(h), \Phi\left(h^{\prime}\right)\right)+120 r r^{\prime}(\text { Theorem 5.3.2 }) \\ & =\frac{5}{3} 6\left(6 \sum_{k=0}^{3} \lambda_{k} \lambda_{k}{ }^{\prime}+3 \sum_{j=1}^{3} \mu_{j} \mu_{j}{ }^{\prime}+4 \nu \nu^{\prime}\right)+120 r r^{\prime} \quad(\text { Theorem 4.6.2 }) \\ & =60 \sum_{k=0}^{3} \lambda_{k} \lambda_{k}{ }^{\prime}+30 \sum_{j=1}^{3} \mu_{j} \mu_{j}{ }^{\prime}+40 \nu \nu^{\prime}+120 r r^{\prime} . \end{aligned}$$</td><td>$$\begin{aligned} B_8(\widetilde{h}, \widetilde{h}') &= \frac{5}{3} B_7(\Phi(h), \Phi(h')) + 120rr' \quad (\text{Theorem } 5.3.2) \\ &= \frac{5}{3} 6\Big(6\sum_{k=0}^{3} \lambda_k {\lambda_k}' + 3\sum_{j=1}^{3} \mu_j {\mu_j}' + 4\nu\nu'\Big) + 120rr' \quad (\text{Theorem } 4.6.2) \\ &= 60\sum_{k=0}^{3} \lambda_k {\lambda_k}' + 30\sum_{j=1}^{3} \mu_j {\mu_j}' + 40\nu\nu' + 120rr'. \end{aligned}$$</td></tr>
<tr><td>0902.0431_EQ0365</td><td>059</td><td>■ 0.264</td><td>\begin{array}{clcccccc} 1 & \longrightarrow & \operatorname{Spin}(8) & \longrightarrow & \left(F_{4} \right)_{E_{1}} & \longrightarrow & S^{8} & \longrightarrow \\ & & \downarrow p^{\prime} & & \downarrow p & & \downarrow= & \\ 1 & \longrightarrow & S O(8) & \longrightarrow & S O(9) & \longrightarrow & S^{8} & \longrightarrow \end{array} .</td><td>$$\begin{array}{clcccccc} 1 & \longrightarrow & \operatorname{Spin}(8) & \longrightarrow & \left(F_{4} \right)_{E_{1}} & \longrightarrow & S^{8} & \longrightarrow \\ & & \downarrow p^{\prime} & & \downarrow p & & \downarrow= & \\ 1 & \longrightarrow & S O(8) & \longrightarrow & S O(9) & \longrightarrow & S^{8} & \longrightarrow \end{array} .$$</td><td>$$\begin{array}{ccccccccc} 1 & \longrightarrow & Spin(8) & \longrightarrow & (F_4)_{E_1} & \longrightarrow & S^8 & \longrightarrow & * \\ & & \downarrow p' & & \downarrow p & & \downarrow = & & \\ 1 & \longrightarrow & SO(8) & \longrightarrow & SO(9) & \longrightarrow & S^8 & \longrightarrow & * \end{array}$$</td></tr>
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0902.0431_EQ0206	035	■ 0.294	`\begin{aligned} & \kappa\left(\alpha_{1}, \alpha_{2}, \alpha_{3}\right)=\left(\kappa \alpha_{1}, \kappa \alpha_{3}, \kappa \alpha_{2}\right), \\ & \pi\left(\alpha_{1}, \alpha_{2}, \alpha_{3}\right)=\left(\kappa \alpha_{3}, \kappa \alpha_{2}, \kappa \alpha_{1}\right), \\ & \nu\left(\alpha_{1}, \alpha_{2}, \alpha_{3}\right)= \left(\alpha_{2}, \alpha_{3}, \alpha_{1}\right), \end{aligned}`	$\begin{aligned} & \kappa\left(\alpha_{1}, \alpha_{2}, \alpha_{3}\right)=\left(\kappa \alpha_{1}, \kappa \alpha_{3}, \kappa \alpha_{2}\right), \\ & \pi\left(\alpha_{1}, \alpha_{2}, \alpha_{3}\right)=\left(\kappa \alpha_{3}, \kappa \alpha_{2}, \kappa \alpha_{1}\right), \\ & \nu\left(\alpha_{1}, \alpha_{2}, \alpha_{3}\right)=\left(\alpha_{2}, \alpha_{3}, \alpha_{1}\right), \end{aligned}$	$\begin{aligned} & \kappa(\alpha_1,\alpha_2,\alpha_3)=(\kappa\alpha_1,\kappa\alpha_3,\kappa\alpha_2), \\ & \pi(\alpha_1,\alpha_2,\alpha_3)=(\kappa\alpha_3,\kappa\alpha_2,\kappa\alpha_1), \\ & \nu(\alpha_1,\alpha_2,\alpha_3)=(\alpha_2,\alpha_3,\alpha_1), \end{aligned}$
0902.0431_EQ0655	108	■ 0.299	`\begin{aligned} & \left(P M \tau h(A, B)^{*}\right)^{*} \left(P N \tau h(A, B)^{*}\right)=\tau h(A, B) M^{*} P^{*} P N \tau h(A, B)^{*} \\ & =\tau\left(h(A, B) \tau\left(M^{*} N\right) h(A, B)^{*}\right), \\ & \frac{\left(P M \tau h(A, B)^{*}\right)\left(h(A, B) Y h(A, B)^{*}\right)}{\left(P M \tau h(A, B)^{*}\right) \times\left(P N \tau h(A, B)^{*}\right)}=\frac{\tau \left(P \tau(M Y) \tau h(A, B)^{*}\right),}{\bar{t} \widetilde{P}(M \times N) \tau^{t}\left(h(A, B)^{*} \right)^{\sim}} \\ & \quad=P(\overline{M \times N}) h(A, B)^{*}=\tau\left(P(\overline{M \times N}) \tau h(A, B)^{*}\right) . \end{aligned}`	$\begin{aligned} & \left(P M \tau h(A, B)^{*}\right)^{*}\left(P N \tau h(A, B)^{*}\right)=\tau h(A, B) M^{*} P^{*} P N \tau h(A, B)^{*} \\ & =\tau\left(h(A, B) \tau\left(M^{*} N\right) h(A, B)^{*}\right), \\ & \frac{\left(P M \tau h(A, B)^{*}\right)\left(h(A, B) Y h(A, B)^{*}\right)}{\left(P M \tau h(A, B)^{*}\right) \times\left(P N \tau h(A, B)^{*}\right)}=\frac{\tau\left(P \tau(M Y) \tau h(A, B)^{*}\right),}{\bar{t} \widetilde{P}(M \times N) \tau^{t}\left(h(A, B)^{*}\right)^{\sim}} \\ & \quad=P(\overline{M \times N}) h(A, B)^{*}=\tau\left(P(\overline{M \times N}) \tau h(A, B)^{*}\right) . \end{aligned}$	$\begin{aligned} (PM\tau h(A,B)^*)^*(PN\tau h(A,B)^*) &= \tau h(A,B)M^*P^*PN\tau h(A,B)^* \\ &= \tau(h(A,B)\tau(M^*N)h(A,B)^*), \\ \frac{(PM\tau h(A,B)^*)(h(A,B)Yh(A,B)^*)}{(PM\tau h(A,B)^*)\times(PN\tau h(A,B)^*)} &= \frac{\tau(P\tau(MY)\tau h(A,B)^*),}{{}^t\widetilde{P}(M\times N)\tau^{\,t}(h(A,B)^*)^{\sim}} \\ &= P(\overline{M\times N})h(A,B)^* = \tau(P(\overline{M\times N})\tau h(A,B)^*). \end{aligned}$
0902.0431_EQ0101	020	■ 0.306	`\begin{aligned} \lambda_{1}-\lambda_{2} & =\alpha_{1}, & \lambda_{1}-\lambda_{3} & =2 \alpha_{1}+3 \alpha_{2}, & \lambda_{2}-\lambda_{3} & =\alpha_{1}+3 \alpha_{2}, \\ \lambda_{1} & =\alpha_{1}+\alpha_{2}, & \lambda_{2} & = \alpha_{2}, & & =\lambda_{3}+2 \alpha_{2} \end{aligned}`	$\begin{aligned} \lambda_{1}-\lambda_{2} & =\alpha_{1}, & \lambda_{1}-\lambda_{3} & =2 \alpha_{1}+3 \alpha_{2}, & \lambda_{2}-\lambda_{3} & =\alpha_{1}+3 \alpha_{2}, \\ \lambda_{1} & =\alpha_{1}+\alpha_{2}, & \lambda_{2} & =\alpha_{2}, & & =\lambda_{3}+2 \alpha_{2} \end{aligned}$	$\begin{aligned} \lambda_1-\lambda_2 &= \alpha_1, & \lambda_1-\lambda_3 &= 2\alpha_1+3\alpha_2, & \lambda_2-\lambda_3 &= \alpha_1+3\alpha_2, \\ \lambda_1 &= \alpha_1+\alpha_2, & \lambda_2 &= \alpha_2, & -\lambda_3 &= \alpha_1+2\alpha_2. \end{aligned}$
0902.0431_EQ1154	195	■ 0.318	`\begin{aligned} & +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^{3} \sum_{j=1}^{3}\left(\boldsymbol{u}_{i} \wedge \boldsymbol{v}_{j} \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge a)=-(*(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}), \boldsymbol{a}) \end{aligned}`	$\begin{aligned} & +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^{3} \sum_{j=1}^{3}\left(\boldsymbol{u}_{i} \wedge \boldsymbol{v}_{j} \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge a)=-(*(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}), \boldsymbol{a}) \end{aligned}$	$\begin{aligned} & +(\boldsymbol{v},\boldsymbol{w})(\boldsymbol{u},\boldsymbol{a})-\sum_{i=1}^{3}\sum_{j=1}^{3}(\boldsymbol{u}_i\wedge\boldsymbol{v}_j\wedge\boldsymbol{v}_{j+1},\boldsymbol{w})(\boldsymbol{u}_{i+1}\wedge\boldsymbol{u}_{i+2}\wedge\boldsymbol{v}_{j+2},\boldsymbol{a}) \\ & =-(\boldsymbol{u}\wedge\boldsymbol{v},\boldsymbol{w}\wedge a)=-(*(*(\boldsymbol{u}\wedge\boldsymbol{v})\wedge\boldsymbol{w}),\boldsymbol{a}). \end{aligned}$

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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ1189	200	0.322	`\begin{aligned} ((\boldsymbol{a} & \times(\boldsymbol{x} \wedge \boldsymbol{y})) \boldsymbol{v}, \boldsymbol{w})=(\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a})-\frac{3}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ & =(\boldsymbol{x}, \boldsymbol{v})(\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a})-(\boldsymbol{y}, \boldsymbol{v})(\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a})+\frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ & =(\boldsymbol{x}, \boldsymbol{v})(*(* \boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w})-(\boldsymbol{y}, \boldsymbol{v})(*(* \boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w})+\frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) . \end{aligned}`	$\begin{aligned} ((\boldsymbol{a} & \times(\boldsymbol{x} \wedge \boldsymbol{y})) \boldsymbol{v}, \boldsymbol{w})=(\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a})-\frac{3}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ & =(\boldsymbol{x}, \boldsymbol{v})(\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a})-(\boldsymbol{y}, \boldsymbol{v})(\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a})+\frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ & =(\boldsymbol{x}, \boldsymbol{v})(*(* \boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w})-(\boldsymbol{y}, \boldsymbol{v})(*(* \boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w})+\frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) . \end{aligned}$	$\begin{aligned} ((\boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y}))\boldsymbol{v}, \boldsymbol{w}) &= (\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a}) - \frac{3}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ &= (\boldsymbol{x}, \boldsymbol{v})(\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a}) - (\boldsymbol{y}, \boldsymbol{v})(\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ &= (\boldsymbol{x}, \boldsymbol{v})(*(*\boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w}) - (\boldsymbol{y}, \boldsymbol{v})(*(*\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$
0902.0431_EQ1157	195	0.323	`\begin{aligned} & 0=D \boldsymbol{u}=\sum_{I} u_{I} D \boldsymbol{e}_{I}=3 \sum_{I \nsupseteq 9} u_{I} \boldsymbol{e}_{I}-6 \sum_{I \ni 9} u_{I} \boldsymbol{e}_{I}, \\ & 0=-{ }^{t} D \boldsymbol{v}=-3 \sum_{J \not \ni 9} v_{J} \boldsymbol{e}_{J}+6 \sum_{J \ni 9} v_{J} \boldsymbol{e}_{J}, \end{aligned}`	$\begin{aligned} & 0=D \boldsymbol{u}=\sum_{I} u_{I} D \boldsymbol{e}_{I}=3 \sum_{I \nsupseteq 9} u_{I} \boldsymbol{e}_{I}-6 \sum_{I \ni 9} u_{I} \boldsymbol{e}_{I}, \\ & 0=-{ }^{t} D \boldsymbol{v}=-3 \sum_{J \not \ni 9} v_{J} \boldsymbol{e}_{J}+6 \sum_{J \ni 9} v_{J} \boldsymbol{e}_{J}, \end{aligned}$	$\begin{aligned} & 0 = D\boldsymbol{u} = \sum_{I} u_I D\boldsymbol{e}_I = 3\sum_{I \not\ni 9} u_I \boldsymbol{e}_I - 6\sum_{I \ni 9} u_I \boldsymbol{e}_I, \\ & 0 = -{}^t D\boldsymbol{v} = -3\sum_{J \not\ni 9} v_J \boldsymbol{e}_J + 6\sum_{J \ni 9} v_J \boldsymbol{e}_J, \end{aligned}$
0902.0431_EQ0460	075	0.325	`\mathfrak{a} \supset\left[\mathfrak{a}, \mathfrak{e}_{6}{ }^{C}\right] \supset\left[\mathfrak{f}_{4}{ }^{C}, \widetilde{\mathfrak{J}}_{0}{ }^{C}\right]= \mathfrak{f}_{4} \widetilde{ }^{C} \mathfrak{J}_{0}{ }^{C} \text { (Lemma 3.2.2) }=\widetilde{ \mathfrak{J}}_{0}^{C} \text { (Proposition 2.4.6.(2)). }`	$\mathfrak{a} \supset\left[\mathfrak{a}, \mathfrak{e}_{6}{ }^{C}\right] \supset\left[\mathfrak{f}_{4}{ }^{C}, \widetilde{\mathfrak{J}}_{0}{ }^{C}\right]= \mathfrak{f}_{4} \widetilde{ }^{C} \mathfrak{J}_{0}{ }^{C} \text { (Lemma 3.2.2) }=\widetilde{ \mathfrak{J}}_{0}^{C} \text { (Proposition 2.4.6.(2)). }$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6{}^C] \supset [\mathfrak{f}_4{}^C, \widetilde{\mathfrak{J}}_0{}^C] = \mathfrak{f}_4 \widetilde{{}^C\mathfrak{J}_0}{}^C \text{ (Lemma 3.2.2) } = \widetilde{\mathfrak{J}}_0{}^C \text{ (Proposition 2.4.6.(2)).}$

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<tr><td>0902.0431_EQ0802</td><td>134</td><td>■ 0.338</td><td><code>\left(\begin{array}{c} \left(\begin{array}{ccc} \cos \left|a_{1}\right| a_{2} \frac{\sin \left|a_{2}\right|}{ \left|a_{2}\right|} a_{3} \frac{\sin \left|a_{3}\right|}{ \left|a_{3}\right|} & 0 & 0 \\ 0 & a_{1} \frac{\sin \left|a_{1}\right|}{\left|a_{1}\right|} \cos \left|a_{2} \right| a_{3} \frac{\sin \left|a_{3}\right|}{\left|a_{3} \right|} & 0 \\ 0 & 0 & a_{1} \frac{\sin \left|a_{1} \right|}{\left|a_{1}\right|} a_{2} \frac{\sin \left|a_{2} \right|}{\left|a_{2}\right|} \cos \left|a_{3}\right| \end{array}\right) \\ \left(\begin{array}{ccc} a_{1} \frac{\sin \left|a_{1}\right|}{\left|a_{1}\right|} \cos \left|a_{2}\right| \cos \left|a_{3}\right| & 0 & 0 \\ 0 & \cos \left|a_{1}\right| a_{2} \frac{\sin \left|a_{2} \right|}{\left|a_{2}\right|} \cos \left|a_{3}\right| & 0 \\ 0 & 0 & \cos \left|a_{1}\right| \cos \left|a_{2} \right| a_{3} \frac{\sin \left|a_{3}\right|}{\left|a_{3} \right|} \end{array}\right) \\ \cos \left|a_{1}\right| \cos \left|a_{2}\right| \cos \left|a_{3}\right| \\ a_{1} \frac{\sin \left|a_{1}\right|}{\left|a_{1}\right|} a_{2} \frac{\sin \left|a_{2}\right|}{\left|a_{2}\right|} a_{3} \frac{\sin \left|a_{3}\right|}{\left|a_{3}\right|} \end{array}\right)</code></td><td>(not rendered)</td><td>$\left(\begin{array}{c} \left(\begin{array}{ccc} \cos \left|a_{1}\right| a_{2} \frac{\sin \left|a_{2}\right|}{\left|a_{2}\right|} a_{3} \frac{\sin \left|a_{3}\right|}{\left|a_{3}\right|} & 0 & 0 \\ 0 & a_{1} \frac{\sin \left|a_{1}\right|}{\left|a_{1}\right|} \cos \left|a_{2}\right| a_{3} \frac{\sin \left|a_{3}\right|}{\left|a_{3}\right|} & 0 \\ 0 & 0 & a_{1} \frac{\sin \left|a_{1}\right|}{\left|a_{1}\right|} a_{2} \frac{\sin \left|a_{2}\right|}{\left|a_{2}\right|} \cos \left|a_{3}\right| \end{array}\right) \\ \left(\begin{array}{ccc} a_{1} \frac{\sin \left|a_{1}\right|}{\left|a_{1}\right|} \cos \left|a_{2}\right| \cos \left|a_{3}\right| & 0 & 0 \\ 0 & \cos \left|a_{1}\right| a_{2} \frac{\sin \left|a_{2}\right|}{\left|a_{2}\right|} \cos \left|a_{3}\right| & 0 \\ 0 & 0 & \cos \left|a_{1}\right| \cos \left|a_{2}\right| a_{3} \frac{\sin \left|a_{3}\right|}{\left|a_{3}\right|} \end{array}\right) \\ \cos \left|a_{1}\right| \cos \left|a_{2}\right| \cos \left|a_{3}\right| \\ a_{1} \frac{\sin \left|a_{1}\right|}{\left|a_{1}\right|} a_{2} \frac{\sin \left|a_{2}\right|}{\left|a_{2}\right|} a_{3} \frac{\sin \left|a_{3}\right|}{\left|a_{3}\right|} \end{array}\right)$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value</td></tr>
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ0875	145	■ 0.339	`\begin{aligned} \varphi_{2}(A) & \left(\left(\begin{array}{ccc} \xi_{1} & x_{3} & \bar{x}_{2} \\ \bar{x}_{3} & \xi_{2} & x_{1} \\ x_{2} & \bar{x}_{1} & \xi_{3} \end{array}\right),\left(\begin{array}{lll} \eta_{1} & y_{3} & \bar{y}_{2} \\ \bar{y}_{3} & \eta_{2} & y_{1} \\ y_{2} & \bar{y}_{1} & \eta_{3} \end{array}\right), \xi, \eta\right) \\ = & \left(\left(\begin{array}{ccc} \xi_{1}^{\prime} & x_{3}^{\prime} & \bar{x}_{2}^{\prime} \\ \bar{x}_{3}^{\prime} & \xi_{2}^{\prime} & x_{1}^{\prime} \\ x_{2}^{\prime} & \bar{x}_{1}^{\prime} & \xi_{3}^{\prime} \end{array} \right),\left(\begin{array}{ccc} \eta_{1}^{\prime} & y_{3}^{\prime} & \bar{y}_{2}^{\prime} \\ \bar{y}_{3}^{\prime} & \eta_{2}^{\prime} & y_{1}^{\prime} \\ y_{2}^{\prime} & \bar{y}_{1}^{\prime} & \eta_{3}^{\prime} \end{array}\right), \xi^{\prime}, \eta^{\prime}\right) \end{aligned}`	$\begin{aligned} \varphi_{2}(A) & \left(\left(\begin{array}{ccc} \xi_{1} & x_{3} & \bar{x}_{2} \\ \bar{x}_{3} & \xi_{2} & x_{1} \\ x_{2} & \bar{x}_{1} & \xi_{3} \end{array}\right),\left(\begin{array}{lll} \eta_{1} & y_{3} & \bar{y}_{2} \\ \bar{y}_{3} & \eta_{2} & y_{1} \\ y_{2} & \bar{y}_{1} & \eta_{3} \end{array}\right), \xi, \eta\right) \\ = & \left(\left(\begin{array}{ccc} \xi_{1}^{\prime} & x_{3}^{\prime} & \bar{x}_{2}^{\prime} \\ \bar{x}_{3}^{\prime} & \xi_{2}^{\prime} & x_{1}^{\prime} \\ x_{2}^{\prime} & \bar{x}_{1}^{\prime} & \xi_{3}^{\prime} \end{array} \right),\left(\begin{array}{ccc} \eta_{1}^{\prime} & y_{3}^{\prime} & \bar{y}_{2}^{\prime} \\ \bar{y}_{3}^{\prime} & \eta_{2}^{\prime} & y_{1}^{\prime} \\ y_{2}^{\prime} & \bar{y}_{1}^{\prime} & \eta_{3}^{\prime} \end{array}\right), \xi^{\prime}, \eta^{\prime}\right) \end{aligned}$	$\begin{aligned} \varphi_2(A)\Big(& \begin{pmatrix} \xi_1 & x_3 & \overline{x}_2 \\ \overline{x}_3 & \xi_2 & x_1 \\ x_2 & \overline{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & y_3 & \overline{y}_2 \\ \overline{y}_3 & \eta_2 & y_1 \\ y_2 & \overline{y}_1 & \eta_3 \end{pmatrix}, \xi, \eta\Big) \\ = \Big(& \begin{pmatrix} {\xi_1}' & {x_3}' & {\overline{x}_2}' \\ {\overline{x}_3}' & {\xi_2}' & {x_1}' \\ {x_2}' & {\overline{x}_1}' & {\xi_3}' \end{pmatrix}, \begin{pmatrix} {\eta_1}' & {y_3}' & {\overline{y}_2}' \\ {\overline{y}_3}' & {\eta_2}' & {y_1}' \\ {y_2}' & {\overline{y}_1}' & {\eta_3}' \end{pmatrix}, \xi', \eta'\Big), \end{aligned}$
0902.0431_EQ0400	066	■ 0.351	`\beta_{1} m=\operatorname{prm} \bar{r}, \quad \beta_{2} m=r m \bar{r} q, \quad \beta_{3} m=\bar{q} r m \bar{r} \bar{p}, \quad m \in \boldsymbol{H} .`	$\beta_{1} m=\operatorname{prm} \bar{r}, \quad \beta_{2} m=r m \bar{r} q, \quad \beta_{3} m=\bar{q} r m \bar{r} \bar{p}, \quad m \in \boldsymbol{H} .$	$\beta_1 m = prm\overline{r}, \quad \beta_2 m = rm\overline{r}q, \quad \beta_3 m = \overline{q}rm\overline{r}\,\overline{p}, \quad m \in \boldsymbol{H}.$
0902.0431_EQ0860	143	■ 0.354	`\begin{array}{cccccccc} 1 & \longrightarrow & \operatorname{Spin}(10) & \longrightarrow & \left(\left(E_{7}\right)^{\kappa, \mu}\right)_{\left(0, E_{1}, 0,1\right)} & \longrightarrow & S^{10} & \longrightarrow \\ & & \downarrow p^{\prime} & & \downarrow p & & \downarrow= & \\ 1 & \longrightarrow & S O(10) & \longrightarrow & S O(11) & \longrightarrow & S^{10} & \longrightarrow \end{array}`	$\begin{array}{cccccccc} 1 & \longrightarrow & \operatorname{Spin}(10) & \longrightarrow & \left(\left(E_{7}\right)^{\kappa, \mu}\right)_{\left(0, E_{1}, 0,1\right)} & \longrightarrow & S^{10} & \longrightarrow \\ & & \downarrow p^{\prime} & & \downarrow p & & \downarrow= & \\ 1 & \longrightarrow & S O(10) & \longrightarrow & S O(11) & \longrightarrow & S^{10} & \longrightarrow \end{array}$	$\begin{array}{ccccccccc} 1 & \longrightarrow & Spin(10) & \longrightarrow & ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} & \longrightarrow & S^{10} & \longrightarrow & * \\ & & \downarrow p' & & \downarrow p & & \downarrow= & & \\ 1 & \longrightarrow & SO(10) & \longrightarrow & SO(11) & \longrightarrow & S^{10} & \longrightarrow & * \end{array}$
0902.0431_EQ0876	145	■ 0.358	`\begin{gathered} \binom{\xi_{1}^{\prime}}{\eta^{\prime}}=A\binom{\xi_{1}}{\eta}, \quad\binom{\xi^{\prime}}{\eta_{1}^{\prime}}=A\binom{\xi}{\eta_{1}}, \quad\binom{\eta_{2}^{\prime}}{\xi_{3}^{\prime}}=A \binom{\eta_{2}}{\xi_{3}}, \quad\binom{\eta_{3}^{\prime}}{\xi_{2}^{\prime}}=A\binom{\eta_{3}}{\xi_{2}} \\ \binom{x_{1}^{\prime}}{y_{1}^{\prime}}=\tau A \binom{x_{1}}{y_{1}}, \quad\binom{x_{2}^{\prime}}{y_{2}^{\prime}}=\binom{x_{2}}{y_{2}}, \quad\binom{x_{3}^{\prime}}{y_{3}^{\prime}}=\binom{x_{3}}{y_{3}} . \end{gathered}`	$\begin{gathered} \binom{\xi_{1}^{\prime}}{\eta^{\prime}}=A\binom{\xi_{1}}{\eta}, \quad\binom{\xi^{\prime}}{\eta_{1}^{\prime}}=A\binom{\xi}{\eta_{1}}, \quad\binom{\eta_{2}^{\prime}}{\xi_{3}^{\prime}}=A \binom{\eta_{2}}{\xi_{3}}, \quad\binom{\eta_{3}^{\prime}}{\xi_{2}^{\prime}}=A\binom{\eta_{3}}{\xi_{2}} \\ \binom{x_{1}^{\prime}}{y_{1}^{\prime}}=\tau A \binom{x_{1}}{y_{1}}, \quad\binom{x_{2}^{\prime}}{y_{2}^{\prime}}=\binom{x_{2}}{y_{2}}, \quad\binom{x_{3}^{\prime}}{y_{3}^{\prime}}=\binom{x_{3}}{y_{3}} . \end{gathered}$	$\begin{gathered} \begin{pmatrix} {\xi_1}' \\ \eta' \end{pmatrix} = A\begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \begin{pmatrix} \xi' \\ {\eta_1}' \end{pmatrix} = A\begin{pmatrix} \xi \\ \eta_1 \end{pmatrix}, \begin{pmatrix} {\eta_2}' \\ {\xi_3}' \end{pmatrix} = A\begin{pmatrix} \eta_2 \\ \xi_3 \end{pmatrix}, \begin{pmatrix} {\eta_3}' \\ {\xi_2}' \end{pmatrix} = A\begin{pmatrix} \eta_3 \\ \xi_2 \end{pmatrix}, \\ \begin{pmatrix} {x_1}' \\ {y_1}' \end{pmatrix} = \tau A\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \begin{pmatrix} {x_2}' \\ {y_2}' \end{pmatrix} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \begin{pmatrix} {x_3}' \\ {y_3}' \end{pmatrix} = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}. \end{gathered}$

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0902.0431_EQ0941	156	■ 0.372	\begin{aligned} & P=\left(F_{1}(1), 0,0,0\right) \\ & \xrightarrow{f} \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{5}+ \boldsymbol{e}_{3} \wedge \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{4} \\ & \xrightarrow{D}\left(D e_{2} \wedge e_{4} \wedge e_{5}+e_{2} \wedge D e_{4} \wedge e_{5}+e_{2} \wedge e_{4} \wedge D e_{5}\right) \\ & +\left(D e_{3} \wedge e_{6} \wedge e_{4}+e_{3} \wedge D e_{6} \wedge e_{4}+e_{3} \wedge e_{6} \wedge D e_{4}\right) \\ & =\left(b_{12} \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{5}+ \left(b_{22}+\frac{\nu}{3}\right) \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{5}- \bar{b}_{23} \boldsymbol{e}_{3} \wedge \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{5}-\bar{l}_{23} \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{5} \right. \\ & +l_{11} \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{5}+l_{31} \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{3} \wedge \boldsymbol{e}_{5}+\left(c_{11}-\frac{\nu}{3}\right) \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{5}-\bar{c}_{13} \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{5} \\ & \left.+l_{12} \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{1}+l_{32} \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{3}+\left(c_{22}-\frac{\nu}{3}\right) \boldsymbol{e}_{3} \wedge \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{4}-\bar{c}_{23} \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{4} \wedge \boldsymbol{e}_{6} \right) \\ & +\left(b_{13} \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{4}+b_{23} \boldsymbol{e}_{2} \wedge \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{4}+\left(b_{33}+\frac{\nu}{3}\right) \boldsymbol{e}_{3} \wedge \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{4}-\bar{l}_{32} \boldsymbol{e}_{5} \wedge \boldsymbol{e}_{6} \wedge \boldsymbol{e}_{4} \right. \\ & +l_{13} \boldsymbol{e}_{3} \wedge \boldsymbol{e}_{1} \wedge \boldsymbol{e}_{4}+l_{23}	$P=(F_1(1),0,0,0)$ $\xrightarrow{f} \boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 + \boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4$ $\xrightarrow{D} (De_2 \wedge e_4 \wedge e_5 + e_2 \wedge De_4 \wedge e_5 + e_2 \wedge e_4 \wedge De_5)$ $+ (De_3 \wedge e_6 \wedge e_4 + e_3 \wedge De_6 \wedge e_4 + e_3 \wedge e_6 \wedge De_4)$ $= \left(b_{12}\boldsymbol{e}_1 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 + \left(b_{22}+\frac{\nu}{3}\right)\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 - \bar{b}_{23}\boldsymbol{e}_3 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 - \bar{l}_{23}\boldsymbol{e}_6 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5\right.$ $+ l_{11}\boldsymbol{e}_2 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_5 + l_{31}\boldsymbol{e}_2 \wedge \boldsymbol{e}_3 \wedge \boldsymbol{e}_5 + \left(c_{11}-\frac{\nu}{3}\right)\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 - \bar{c}_{13}\boldsymbol{e}_2 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_5$ $\left.+l_{12}\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_1 + l_{32}\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_3 + \left(c_{22}-\frac{\nu}{3}\right)\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 - \bar{c}_{23}\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_6\right)$ $+\left(b_{13}\boldsymbol{e}_1 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 + b_{23}\boldsymbol{e}_2 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 + \left(b_{33}+\frac{\nu}{3}\right)\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 - \bar{l}_{32}\boldsymbol{e}_5 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4\right.$ $+ l_{13}\boldsymbol{e}_3 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_4 + l_{23}\boldsymbol{e}_3 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_4 + c_{23}\boldsymbol{e}_3 \wedge \boldsymbol{e}_5 \wedge \boldsymbol{e}_4 + \left(c_{33}-\frac{\nu}{3}\right)\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4$ $\left.+l_{11}e_3 \wedge e_6 \wedge e_1 + l_{21}e_3 \wedge e_6 \wedge e_2 + \left(c_{11}-\frac{\nu}{3}\right)e_3 \wedge e_6 \wedge e_4 - \bar{c}_{12}e_3 \wedge e_6 \wedge e_5\right)$ $\xrightarrow{f^{-1}} \left(\begin{array}{ccc} 0 & h\left(b_{13}, c_{13}\right) & 0 \\ * & h\left(b_{23}+\bar{c}_{23}\right) & h\left(b_{22}-c_{33}-\frac{\nu}{3}, -b_{33}+c_{33}+\frac{\nu}{3}\right) \\ * & & -h\left(\bar{b}_{23}+c_{23}\right) \end{array}\right)$	$P=(F_1(1),0,0,0)$ $\xrightarrow{f} \boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 + \boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4$ $\xrightarrow{D} (D\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 + \boldsymbol{e}_2 \wedge D\boldsymbol{e}_4 \wedge \boldsymbol{e}_5 + \boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge D\boldsymbol{e}_5)$ $+(D\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 + \boldsymbol{e}_3 \wedge D\boldsymbol{e}_6 \wedge \boldsymbol{e}_4 + \boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge D\boldsymbol{e}_4)$ $= \Big(b_{12}\boldsymbol{e}_1 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 + \Big(b_{22}+\frac{\nu}{3}\Big)\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 - \overline{b}_{23}\boldsymbol{e}_3 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 - \overline{l}_{23}\boldsymbol{e}_6 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5$ $+l_{11}\boldsymbol{e}_2 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_5 + l_{31}\boldsymbol{e}_2 \wedge \boldsymbol{e}_3 \wedge \boldsymbol{e}_5 + \Big(c_{11}-\frac{\nu}{3}\Big)\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 - \overline{c}_{13}\boldsymbol{e}_2 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_5$ $+l_{12}\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_1 + l_{32}\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_3 + \Big(c_{22}-\frac{\nu}{3}\Big)\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 - \overline{c}_{23}\boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_6\Big)$ $+\Big(b_{13}\boldsymbol{e}_1 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 + b_{23}\boldsymbol{e}_2 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 + \Big(b_{33}+\frac{\nu}{3}\Big)\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 - \overline{l}_{32}\boldsymbol{e}_5 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4$ $+l_{13}\boldsymbol{e}_3 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_4 + l_{23}\boldsymbol{e}_3 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_4 + c_{23}\boldsymbol{e}_3 \wedge \boldsymbol{e}_5 \wedge \boldsymbol{e}_4 + \Big(c_{33}-\frac{\nu}{3}\Big)\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4$ $+l_{11}\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_1 + l_{21}\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_2 + \Big(c_{11}-\frac{\nu}{3}\Big)\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 - \overline{c}_{12}\boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_5\Big)$ $\xrightarrow{f^{-1}} \left(\begin{array}{c} \left(\begin{array}{ccc} 0 & h(b_{13},c_{13}) & 0 \\ * & h(b_{23}+\overline{c}_{23}) & h\left(b_{22}-c_{33}-\frac{\nu}{3}, -b_{33}+c_{33}+\frac{\nu}{3}\right) \\ h(\overline{c}_{12},\overline{b}_{12}) & * & -h(\overline{b}_{23}+c_{23}) \end{array}\right) \\ \left(\begin{array}{ccc} -h(l_{23}+l_{32}) & h(l_{13},\overline{l}_{31}) & * \\ * & 0 & -h(l_{11}) \\ h(l_{21},\overline{l}_{12}) & * & 0 \end{array}\right) \\ 0 \\ -h(\overline{l}_{23}+\overline{l}_{32}) \end{array}\right)$

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0902.0431_EQ0093	019	■ 0.382	`\begin{aligned} H_{1} H_{1} e_{2} & =H_{1} e_{3}=-e_{2}, \quad \end{aligned}`	$H_1 H_1 e_2 = H_1 e_3 = -e_2,$	$H_1H_1e_2 = H_1e_3 = -e_2, \quad H_1H_1e_3 = -H_1e_2 = -e_3,$ $H_1H_1e_4 = -H_1e_5 = -e_4, \quad H_1H_1e_5 = H_1e_4 = -e,$ $H_1H_1e_i = 0$ otherwise.
0902.0431_EQ0371	061	■ 0.388	`\begin{aligned} & \operatorname{tr}(X)=\operatorname{tr}(\alpha X)(\text { Lemma 2.2.1.(2) })=\sum_{i=1}^{3} \xi_{i}, \\ & (X, X)=(\alpha X, \alpha X)(\text { Lemma 2.2.4 })=\sum_{i=1}^{3} \xi_{i}{ }^{2}, \\ & \operatorname{tr}(X, X, X)=\operatorname{tr}(\alpha X, \alpha X, \alpha X)(\text { Lemma 2.2.4 })=\sum_{i=1}^{3} \xi_{i}{ }^{3} . \end{aligned}`	$\operatorname{tr}(X)=\operatorname{tr}(\alpha X)(\text{ Lemma 2.2.1.(2) })=\sum_{i=1}^{3} \xi_{i},$ $(X, X)=(\alpha X, \alpha X)(\text{ Lemma 2.2.4 })=\sum_{i=1}^{3} \xi_{i}{}^{2},$ $\operatorname{tr}(X, X, X)=\operatorname{tr}(\alpha X, \alpha X, \alpha X)(\text{ Lemma 2.2.4 })=\sum_{i=1}^{3} \xi_{i}{}^{3}.$	$\operatorname{tr}(X) = \operatorname{tr}(\alpha X)$ (Lemma 2.2.1.(2)) $= \sum_{i=1}^3 \xi_i,$ $(X, X) = (\alpha X, \alpha X)$ (Lemma 2.2.4) $= \sum_{i=1}^3 \xi_i{}^2,$ $\operatorname{tr}(X, X, X) = \operatorname{tr}(\alpha X, \alpha X, \alpha X)$ (Lemma 2.2.4) $= \sum_{i=1}^3 \xi_i{}^3.$
0902.0431_EQ0602	100	■ 0.396	`\left.\left.\begin{array}{rl} \left(\mathfrak{J}^{C} \right)_{\tau \gamma} & =\left\{X \in \mathfrak{J}^{C} \mid \tau \gamma X=X\right\} \\ & =\left\{\left(\begin{array}{ccc} \mu_{1} & m_{3} & \bar{m}_{2} \\ \bar{m}_{3} & \mu_{2} & m_{1} \\ m_{2} & \bar{m}_{1} & \mu_{3} \end{array}\right)+i\left(\begin{array}{ccc} 0 & a_{3} e_{4} & -a_{2} e_{4} \\ -a_{3} e_{4} & 0 & a_{1} e_{4} \\ a_{2} e_{4} & -a_{1} e_{4} & 0 \end{array}\right)\right. \end{array} \right\rvert\, \begin{array}{l} \mu_{i} \in \boldsymbol{R} \\ m_{i}, a_{i} \in \boldsymbol{H} \end{array}\right\}`	$\left.\left.\begin{array}{rl} \left(\mathfrak{J}^{C}\right)_{\tau \gamma} & =\left\{X \in \mathfrak{J}^{C} \mid \tau \gamma X=X\right\} \\ & =\left\{\left(\begin{array}{ccc} \mu_{1} & m_{3} & \bar{m}_{2} \\ \bar{m}_{3} & \mu_{2} & m_{1} \\ m_{2} & \bar{m}_{1} & \mu_{3} \end{array}\right)+i\left(\begin{array}{ccc} 0 & a_{3} e_{4} & -a_{2} e_{4} \\ -a_{3} e_{4} & 0 & a_{1} e_{4} \\ a_{2} e_{4} & -a_{1} e_{4} & 0 \end{array}\right)\right. \end{array} \right\rvert\, \begin{array}{l} \mu_{i} \in \boldsymbol{R} \\ m_{i}, a_{i} \in \boldsymbol{H} \end{array}\right\}$	$(\mathfrak{J}^C)_{\tau\gamma} = \{X \in \mathfrak{J}^C \mid \tau\gamma X = X\}$ $= \left\{ \left(\begin{array}{ccc} \mu_1 & m_3 & \overline{m}_2 \\ \overline{m}_3 & \mu_2 & m_1 \\ m_2 & \overline{m}_1 & \mu_3 \end{array}\right) + i \left(\begin{array}{ccc} 0 & a_3e_4 & -a_2e_4 \\ -a_3e_4 & 0 & a_1e_4 \\ a_2e_4 & -a_1e_4 & 0 \end{array}\right) \,\middle\vert\, \begin{array}{l} \mu_i \in \boldsymbol{R} \\ m_i, a_i \in \boldsymbol{H} \end{array} \right\}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ0900	149	■ 0.397	`\begin{aligned} P= & (X, Y, \xi, \eta) \\ \stackrel{ \chi}{\longrightarrow} & k\left(g X-\frac{\xi}{2} E \right) J+e_{1} k\left(g(\gamma Y)-\frac{\eta}{2} E \right) J \\ \longrightarrow & e_{1} k(T) k\left(g X-\frac{\xi}{2} E\right) J-k(T) k\left(g(\gamma Y)-\frac{ \eta}{2} E\right) J \\ & +e_{1} k\left(g X-\frac{\xi}{2} E\right) J^{t} k(T)-k\left(g(\gamma Y)-\frac{\eta}{2} E\right) J^{t} k(T) \\ = & k(-T g(\gamma Y)-g(\gamma Y) T+\eta T) J+e_{1} k(T(g X)+(g X) T-\xi T) J \\ = & k(-2 g A \circ g(\gamma Y)+\eta g A) J+e_{1} k(2 g A \circ g X-\xi g A) J \\ = & k\left(-2 g(\gamma A \times Y)-\frac{1}{2}(A, Y) E+\eta g A\right) J \\ & +e_{1} k \left(2 g(\gamma A \times \gamma X)+\frac{1}{2}(\gamma A, X) E-\xi g A\right) J(\text { Lemma } 3.12 .1) \\ = & \chi\left(\begin{array}{cccc} -2 \gamma A \times Y+ \eta A \\ 2 A \times X-\xi \gamma A \\ (A, Y) \end{array} \right)=\chi\left(\left(\begin{array}{cccc} 0 & -2 \gamma A & 0 & A \\ 2 A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right) \left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array} \right)\right) \\ = & \chi(\Phi(0, A,-\gamma A, 0) P) \end{aligned}`	$\begin{aligned} P= & (X, Y, \xi, \eta) \\ \stackrel{\chi}{\longrightarrow} & k\left(g X-\frac{\xi}{2} E\right) J+e_{1} k\left(g(\gamma Y)-\frac{\eta}{2} E\right) J \\ \longrightarrow & e_{1} k(T) k\left(g X-\frac{\xi}{2} E\right) J-k(T) k\left(g(\gamma Y)-\frac{\eta}{2} E\right) J \\ & +e_{1} k\left(g X-\frac{\xi}{2} E\right) J^{t} k(T)-k\left(g(\gamma Y)-\frac{\eta}{2} E\right) J^{t} k(T) \\ = & k(-T g(\gamma Y)-g(\gamma Y) T+\eta T) J+e_{1} k(T(g X)+(g X) T-\xi T) J \\ = & k(-2 g A \circ g(\gamma Y)+\eta g A) J+e_{1} k(2 g A \circ g X-\xi g A) J \\ = & k\left(-2 g(\gamma A \times Y)-\frac{1}{2}(A, Y) E+\eta g A\right) J \\ & +e_{1} k\left(2 g(\gamma A \times \gamma X)+\frac{1}{2}(\gamma A, X) E-\xi g A\right) J(\text { Lemma } 3.12.1) \\ = & \chi\left(\begin{array}{c} -2 \gamma A \times Y+\eta A \\ 2 A \times X-\xi \gamma A \\ (A, Y) \end{array}\right)=\chi\left(\left(\begin{array}{cccc} 0 & -2 \gamma A & 0 & A \\ 2 A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array}\right)\right) \\ = & \chi(\Phi(0, A,-\gamma A, 0) P) \end{aligned}$	$\begin{aligned} P &= (X, Y, \xi, \eta) \\ &\xrightarrow{\chi} k\Big(gX-\frac{\xi}{2}E\Big)J+e_1 k\Big(g(\gamma Y)-\frac{\eta}{2}E\Big)J \\ &\longrightarrow e_1 k(T) k\Big(gX-\frac{\xi}{2}E\Big)J-k(T)k\Big(g(\gamma Y)-\frac{\eta}{2}E\Big)J \\ &\quad +e_1 k\Big(gX-\frac{\xi}{2}E\Big)J^{t}k(T)-k\Big(g(\gamma Y)-\frac{\eta}{2}E\Big)J^{t}k(T) \\ &= k(-Tg(\gamma Y)-g(\gamma Y)T+\eta T)J+e_1 k(T(gX)+(gX)T-\xi T)J \\ &= k(-2gA\circ g(\gamma Y)+\eta gA)J+e_1 k(2gA\circ gX-\xi gA)J \\ &= k\Big(-2g(\gamma A\times Y)-\frac{1}{2}(A,Y)E+\eta gA\Big)J \\ &\quad +e_1 k\Big(2g(\gamma A\times\gamma X)+\frac{1}{2}(\gamma A,X)E-\xi gA\Big)J \text{ (Lemma 3.12.1)} \\ &= \chi\begin{pmatrix} -2\gamma A\times Y+\eta A \\ 2A\times X-\xi\gamma A \\ (A,Y) \\ (-\gamma A,X) \end{pmatrix}=\chi\left(\begin{pmatrix} 0 & -2\gamma A & 0 & A \\ 2A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} X \\ Y \\ \xi \\ \eta \end{pmatrix}\right) \\ &= \chi(\Phi(0,A,-\gamma A,0)P). \end{aligned}$
0902.0431_EQ1080	183	■ 0.403	`S_{1} \tau S_{2}-S_{2} \tau S_{1}-\frac{1}{8}\left(S_{1} \tau S_{2}-S_{2} \tau S_{1}\right) \in \mathfrak{s u}(8),`	$S_{1} \tau S_{2}-S_{2} \tau S_{1}-\frac{1}{8}\left(S_{1} \tau S_{2}-S_{2} \tau S_{1}\right) \in \mathfrak{su}(8),$	$S_1\tau S_2-S_2\tau S_1-\frac{1}{8}(S_1\tau S_2-S_2\tau S_1)\in\mathfrak{su}(8),$
0902.0431_EQ1030	173	■ 0.410	`\begin{aligned} & \lambda_{0}+\frac{1}{2} \mu_{1}-\frac{2}{3} \nu=2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \\ & \lambda_{0}-\frac{1}{2} \mu_{1}+\frac{2}{3} \nu=2 \quad 3 \quad 5 \quad 4 \\ & \lambda_{1}+\frac{1}{2} \mu_{1}-\frac{2}{3} \nu=0 \quad 1 \quad 1 \quad 1 \quad 2 \\ & \lambda_{1}-\frac{1}{2} \mu_{1}+\frac{2}{3} \nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}`	$\begin{aligned} & \lambda_{0}+\frac{1}{2} \mu_{1}-\frac{2}{3} \nu=2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \\ & \lambda_{0}-\frac{1}{2} \mu_{1}+\frac{2}{3} \nu=2 \quad 3 \quad 5 \quad 4 \\ & \lambda_{1}+\frac{1}{2} \mu_{1}-\frac{2}{3} \nu=0 \quad 1 \quad 1 \quad 1 \quad 2 \\ & \lambda_{1}-\frac{1}{2} \mu_{1}+\frac{2}{3} \nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}$	$\begin{aligned} & \lambda_0+\frac{1}{2}\mu_1-\frac{2}{3}\nu=2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \lambda_0-\frac{1}{2}\mu_1+\frac{2}{3}\nu=2 \quad 3 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 3 \\ & \lambda_1+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_1-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ0492	081	■ 0.411	`\delta\left(\begin{array}{ccc} \xi_{1} & x_{12} & x_{13} \\ x_{21} & \xi_{2} & x_{23} \\ x_{31} & x_{32} & \xi_{3} \end{array}\right)=\left(\begin{array}{ccc} 0 & \delta_{3} x_{12} & \bar{\delta}_{2} \bar{x}_{13} \\ \bar{\delta}_{3} \bar{x}_{21} & 0 & \delta_{1} x_{23} \\ \delta_{2} x_{31} & \bar{\delta}_{1} \bar{x}_{32} & 0 \end{array}\right) .`	$\delta\left(\begin{array}{ccc} \xi_{1} & x_{12} & x_{13} \\ x_{21} & \xi_{2} & x_{23} \\ x_{31} & x_{32} & \xi_{3} \end{array}\right)=\left(\begin{array}{ccc} 0 & \delta_{3} x_{12} & \bar{\delta}_{2} \bar{x}_{13} \\ \bar{\delta}_{3} \bar{x}_{21} & 0 & \delta_{1} x_{23} \\ \delta_{2} x_{31} & \bar{\delta}_{1} \bar{x}_{32} & 0 \end{array}\right) .$	$\delta\left(\begin{array}{ccc} \xi_{1} & x_{12} & x_{13} \\ x_{21} & \xi_{2} & x_{23} \\ x_{31} & x_{32} & \xi_{3} \end{array}\right)=\left(\begin{array}{ccc} 0 & \delta_{3} x_{12} & \overline{\delta_{2} x_{13}} \\ \overline{\delta_{3} x_{21}} & 0 & \delta_{1} x_{23} \\ \delta_{2} x_{31} & \overline{\delta_{1} x_{32}} & 0 \end{array}\right) .$
0902.0431_EQ1097	186	■ 0.420	`\begin{aligned} \zeta[(i c E), l(S) I]=\zeta(2 l(i c S) I) \\ \quad=\Theta\left(0,4 \lambda \gamma \chi^{-1}(i c S), 4 \chi^{-1}(i c S), 0,0,0\right) \\ \quad=\Theta\left(0,4 \chi^{-1}(c S),-4 \lambda \gamma \chi^{-1}(c S), 0,0,0\right) \quad(\text { Lemma 5.8.3 }) \\ \quad=\left[\Theta(0,0,0,0,2 c,-2 c), \Theta\left(0,2 \lambda \gamma \chi^{-1} S, 2 \chi^{-1} S, 0,0,0\right)\right] \\ \quad=[\zeta l(i c E), \zeta(l(S) I)] . \end{aligned}`	$\begin{aligned} \zeta[(i c E), l(S) I]=\zeta(2 l(i c S) I) \\ \quad=\Theta\left(0,4 \lambda \gamma \chi^{-1}(i c S), 4 \chi^{-1}(i c S), 0,0,0\right) \\ \quad=\Theta\left(0,4 \chi^{-1}(c S),-4 \lambda \gamma \chi^{-1}(c S), 0,0,0\right) \quad(\text { Lemma 5.8.3 }) \\ \quad=\left[\Theta(0,0,0,0,2 c,-2 c), \Theta\left(0,2 \lambda \gamma \chi^{-1} S, 2 \chi^{-1} S, 0,0,0\right)\right] \\ \quad=[\zeta l(i c E), \zeta(l(S) I)] . \end{aligned}$	$\begin{aligned} &\zeta[(icE), l(S)I]=\zeta(2l(icS)I) \\ &\quad=\Theta(0,4\lambda\gamma\chi^{-1}(icS),4\chi^{-1}(icS),0,0,0) \\ &\quad=\Theta(0,4\chi^{-1}(cS),-4\lambda\gamma\chi^{-1}(cS),0,0,0) \quad (\text{Lemma 5.8.3}) \\ &\quad=[\Theta(0,0,0,0,2c,-2c),\Theta(0,2\lambda\gamma\chi^{-1}S,2\chi^{-1}S,0,0,0)] \\ &\quad=[\zeta l(icE),\zeta(l(S)I)]. \end{aligned}$
0902.0431_EQ0694	115	■ 0.434	`\mathfrak{e}_{7}^{C} \ni f_{r} \Phi f_{r}{ }^{-1}=\left(\begin{array}{cccc} g & r^{-1} l & r^{-2} C & r A \\ r k & h & r^{-1} B & r^{2} D \\ r^{2} c & r a & \nu & r^{3} \lambda \\ r^{-1} b & r^{-2} d & r^{-3} k & \mu \end{array}\right)`	$\mathfrak{e}_{7}^{C} \ni f_{r} \Phi f_{r}{ }^{-1}=\left(\begin{array}{cccc} g & r^{-1} l & r^{-2} C & r A \\ r k & h & r^{-1} B & r^{2} D \\ r^{2} c & r a & \nu & r^{3} \lambda \\ r^{-1} b & r^{-2} d & r^{-3} k & \mu \end{array}\right)$	$\mathfrak{e}_{7}{}^{C} \ni f_{r} \varPhi f_{r}{}^{-1}=\left(\begin{array}{cccc} g & r^{-1} l & r^{-2} C & r A \\ r k & h & r^{-1} B & r^{2} D \\ r^{2} c & r a & \nu & r^{3} \lambda \\ r^{-1} b & r^{-2} d & r^{-3} k & \mu \end{array}\right)$
0902.0431_EQ0642	106	■ 0.434	`\begin{aligned} & h(a, b)+h\left(a^{\prime}, b^{\prime}\right)=h\left(a+a^{\prime}, b+b^{\prime}\right), \quad h(c a, c b)=c h(a, b), c \in \boldsymbol{C} \\ & h(A, B)+h\left(A^{\prime}, B^{\prime}\right)=h\left(A+A^{\prime}, B+B^{\prime}\right), \quad h(c A, c B)=h(A, B), c \in C . \end{aligned}`	$\begin{aligned} & h(a, b)+h\left(a^{\prime}, b^{\prime}\right)=h\left(a+a^{\prime}, b+b^{\prime}\right), \quad h(c a, c b)=c h(a, b), c \in \boldsymbol{C} \\ & h(A, B)+h\left(A^{\prime}, B^{\prime}\right)=h\left(A+A^{\prime}, B+B^{\prime}\right), \quad h(c A, c B)=h(A, B), c \in C . \end{aligned}$	$\begin{aligned} & h(a,b)+h(a',b')=h(a+a',b+b'), \quad h(ca,cb)=ch(a,b), c \in \boldsymbol{C}, \\ & h(A,B)+h(A',B')=h(A+A',B+B'), \quad h(cA,cB)=h(A,B), c \in \boldsymbol{C}. \end{aligned}$
0902.0431_EQ0558	094	■ 0.435	`\begin{array}{clccccccc} 1 & \longrightarrow & S p i n(9) & \longrightarrow & \left(E_{6}\right)_{E_{1}} & \longrightarrow & S^{9} & \longrightarrow & * \\ & & \downarrow p^{\prime} & & \downarrow p & & \downarrow= & & \\ 1 & \longrightarrow & S O(9) & \longrightarrow & S O(10) & \longrightarrow & S^{9} & \longrightarrow & * \end{array}`	$\begin{array}{clccccccc} 1 & \longrightarrow & S p i n(9) & \longrightarrow & \left(E_{6}\right)_{E_{1}} & \longrightarrow & S^{9} & \longrightarrow & * \\ & & \downarrow p^{\prime} & & \downarrow p & & \downarrow= & & \\ 1 & \longrightarrow & S O(9) & \longrightarrow & S O(10) & \longrightarrow & S^{9} & \longrightarrow & * \end{array}$	$\begin{array}{ccccccccc} 1 & \longrightarrow & Spin(9) & \longrightarrow & (E_{6})_{E_{1}} & \longrightarrow & S^{9} & \longrightarrow & * \\ & & \downarrow p' & & \downarrow p & & \downarrow= & & \\ 1 & \longrightarrow & SO(9) & \longrightarrow & SO(10) & \longrightarrow & S^{9} & \longrightarrow & * \end{array}$

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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ0332	053	■ 0.438	`\begin{aligned} \pm\left(\lambda_{k}-\lambda_{l} \right), & \pm\left(\lambda_{k}+\lambda_{l}\right), \quad 0 \leq k<l \leq 3, \\ \pm \lambda_{0}, \quad \pm \lambda_{1}, \quad & \pm \lambda_{2}, \quad \pm \lambda_{3}, \\ \pm \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}+\lambda_{2}-\lambda_{3}\right), & \pm \frac{1}{2}\left(\lambda_{0}+\lambda_{1}+\lambda_{2}-\lambda_{3}\right), \\ \pm \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}+\lambda_{2}+\lambda_{3}\right), & \pm \frac{1}{2}\left(\lambda_{0}-\lambda_{1}+\lambda_{2}+\lambda_{3}\right), \\ \pm \frac{1}{2}\left(\lambda_{0}-\lambda_{1}+\lambda_{2}-\lambda_{3}\right), & \pm \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}+\lambda_{2}-\lambda_{3}\right), \\ \pm \frac{1}{2}\left(\lambda_{0}+\lambda_{1}+\lambda_{2}+\lambda_{3}\right), & \pm \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}+\lambda_{2}+\lambda_{3}\right) . \end{aligned}`	$\begin{aligned} \pm\left(\lambda_{k}-\lambda_{l} \right), & \pm\left(\lambda_{k}+\lambda_{l}\right), \quad 0 \leq k<l \leq 3, \\ \pm \lambda_{0}, \quad \pm \lambda_{1}, \quad & \pm \lambda_{2}, \quad \pm \lambda_{3}, \\ \pm \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}+\lambda_{2}-\lambda_{3}\right), & \pm \frac{1}{2}\left(\lambda_{0}+\lambda_{1}+\lambda_{2}-\lambda_{3}\right), \\ \pm \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}+\lambda_{2}+\lambda_{3}\right), & \pm \frac{1}{2}\left(\lambda_{0}-\lambda_{1}+\lambda_{2}+\lambda_{3}\right), \\ \pm \frac{1}{2}\left(\lambda_{0}-\lambda_{1}+\lambda_{2}-\lambda_{3}\right), & \pm \frac{1}{2}\left(-\lambda_{0}+\lambda_{1}+\lambda_{2}-\lambda_{3}\right), \\ \pm \frac{1}{2}\left(\lambda_{0}+\lambda_{1}+\lambda_{2}+\lambda_{3}\right), & \pm \frac{1}{2}\left(-\lambda_{0}-\lambda_{1}+\lambda_{2}+\lambda_{3}\right) . \end{aligned}$	$\begin{gathered} \pm(\lambda_k-\lambda_l), \quad \pm(\lambda_k+\lambda_l)\ , \quad 0 \leq k < l \leq 3, \\ \pm\lambda_0, \quad \pm\lambda_1, \quad \pm\lambda_2, \quad \pm\lambda_3, \end{gathered}$ $\begin{aligned} &\pm\frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2-\lambda_3), && \pm\frac{1}{2}(\ \ \lambda_0+\lambda_1+\lambda_2-\lambda_3), \\ &\pm\frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2+\lambda_3), && \pm\frac{1}{2}(\ \ \lambda_0-\lambda_1+\lambda_2+\lambda_3), \\ &\pm\frac{1}{2}(\ \ \lambda_0-\lambda_1+\lambda_2-\lambda_3), && \pm\frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3), \\ &\pm\frac{1}{2}(\ \ \lambda_0+\lambda_1+\lambda_2+\lambda_3), && \pm\frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3). \end{aligned}$
0902.0431_EQ1211	203	■ 0.448	`\begin{aligned} \exp (\operatorname{ad} & (Z, 0))(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) \\ = & (\exp (\operatorname{ad}(Z)) C, D,((\exp Z) \boldsymbol{x}) \otimes \boldsymbol{a}, \\ & ((\exp Z) \boldsymbol{b}) \otimes \boldsymbol{y},\left(\left(\exp \left(-{ }^{t} Z\right)\right) \boldsymbol{c} \otimes \boldsymbol{z},\left(\left(\exp \left(-{ }^{t} Z\right) \right) \boldsymbol{w}\right) \otimes \boldsymbol{d} \right) \\ = & (\operatorname{Ad}(\exp Z) C, D,((\exp Z) \boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \left.((\exp Z) \boldsymbol{b}) \otimes \boldsymbol{y}, \left({ }^{t}(\exp Z)^{-1} \boldsymbol{c}\right) \otimes \boldsymbol{z},\left({ }^{t}(\exp Z)^{-1} \boldsymbol{w}\right) \otimes \boldsymbol{d}\right) \\ = & \varphi_{1}(\exp Z)(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) . \end{aligned}`	$\begin{aligned} \exp (\operatorname{ad} & (Z, 0))(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) \\ = & (\exp (\operatorname{ad}(Z)) C, D,((\exp Z) \boldsymbol{x}) \otimes \boldsymbol{a}, \\ & ((\exp Z) \boldsymbol{b}) \otimes \boldsymbol{y},\left(\left(\exp \left(-{ }^{t} Z\right)\right) \boldsymbol{c} \otimes \boldsymbol{z},\left(\left(\exp \left(-{ }^{t} Z\right) \right) \boldsymbol{w}\right) \otimes \boldsymbol{d} \right) \\ = & (\operatorname{Ad}(\exp Z) C, D,((\exp Z) \boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \left.((\exp Z) \boldsymbol{b}) \otimes \boldsymbol{y}, \left({ }^{t}(\exp Z)^{-1} \boldsymbol{c}\right) \otimes \boldsymbol{z},\left({ }^{t}(\exp Z)^{-1} \boldsymbol{w}\right) \otimes \boldsymbol{d}\right) \\ = & \varphi_{1}(\exp Z)(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) . \end{aligned}$	$\begin{aligned} &\exp(\mathrm{ad}(Z,0))(C,D,\boldsymbol{x}\otimes\boldsymbol{a},\boldsymbol{b}\otimes\boldsymbol{y},\boldsymbol{c}\otimes\boldsymbol{z},\boldsymbol{w}\otimes\boldsymbol{d}) \\ &\quad = (\exp(\mathrm{ad}(Z))C, D, ((\exp Z)\boldsymbol{x})\otimes\boldsymbol{a}, \\ &\qquad ((\exp Z)\boldsymbol{b})\otimes\boldsymbol{y}, ((\exp(-{}^tZ))\boldsymbol{c}\otimes\boldsymbol{z}, ((\exp(-{}^tZ))\boldsymbol{w})\otimes\boldsymbol{d}) \\ &\quad = (\mathrm{Ad}(\exp Z)C, D, ((\exp Z)\boldsymbol{x})\otimes\boldsymbol{a}, \\ &\qquad ((\exp Z)\boldsymbol{b})\otimes\boldsymbol{y}, ({}^t(\exp Z)^{-1}\boldsymbol{c})\otimes\boldsymbol{z}, ({}^t(\exp Z)^{-1}\boldsymbol{w})\otimes\boldsymbol{d}) \\ &\quad = \varphi_1(\exp Z)(C,D,\boldsymbol{x}\otimes\boldsymbol{a},\boldsymbol{b}\otimes\boldsymbol{y},\boldsymbol{c}\otimes\boldsymbol{z},\boldsymbol{w}\otimes\boldsymbol{d}). \end{aligned}$

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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ1039	177	■ 0.452	\begin{aligned} H_{\alpha_{3}} & =\left(0,0,0, \frac{1}{60}, 0,0\right), \\ H_{\alpha_{4}} & =\left(\Phi\left(\frac{1}{90}\left(-E_{1}+2 E_{2}-E_{3} \right)^{\sim}, 0,0,-\frac{1}{120}\right), 0,0,- \frac{1}{120}, 0,0\right), \\ H_{\alpha_{5}} & =\left(\Phi\left(\frac{1}{60}\left(H_{3}-\left(E_{2}-E_{3} \right)^{\sim}\right), 0,0,0\right), 0,0,0,0,0\right), \\ H_{\alpha_{6}} & =\left(\Phi\left(\frac{1}{60} \left(H_{2}-H_{3}\right), 0,0,0\right), 0,0,0,0,0\right), \\ H_{\alpha_{7}} & =\left(\Phi\left(\frac{1}{60} \left(H_{1}-H_{2}\right), 0,0,0\right), 0,0,0,0,0\right), \\ H_{\alpha_{8}} & =\left(\Phi\left(0,0,0, \frac{1}{40} \right), 0,0,-\frac{1}{120}, 0,0\right) . \end{aligned}	$\begin{aligned} H_{\alpha_{3}} & =\left(0,0,0, \frac{1}{60}, 0,0\right), \\ H_{\alpha_{4}} & =\left(\Phi\left(\frac{1}{90}\left(-E_{1}+2 E_{2}-E_{3} \right)^{\sim}, 0,0,-\frac{1}{120}\right), 0,0,- \frac{1}{120}, 0,0\right), \\ H_{\alpha_{5}} & =\left(\Phi\left(\frac{1}{60}\left(H_{3}-\left(E_{2}-E_{3} \right)^{\sim}\right), 0,0,0\right), 0,0,0,0,0\right), \\ H_{\alpha_{6}} & =\left(\Phi\left(\frac{1}{60} \left(H_{2}-H_{3}\right), 0,0,0\right), 0,0,0,0,0\right), \\ H_{\alpha_{7}} & =\left(\Phi\left(\frac{1}{60} \left(H_{1}-H_{2}\right), 0,0,0\right), 0,0,0,0,0\right), \\ H_{\alpha_{8}} & =\left(\Phi\left(0,0,0, \frac{1}{40} \right), 0,0,-\frac{1}{120}, 0,0\right) . \end{aligned}$	$\begin{aligned} H_{\alpha_3} &= \Big(0,0,0,\frac{1}{60},0,0\Big), \\ H_{\alpha_4} &= \Big(\varPhi\Big(\frac{1}{90}(-E_1+2E_2-E_3)^{\sim},0,0,-\frac{1}{120}\Big),0,0,-\frac{1}{120},0,0\Big), \\ H_{\alpha_5} &= \Big(\varPhi\Big(\frac{1}{60}(H_3-(E_2-E_3)^{\sim}),0,0,0\Big),0,0,0,0,0\Big), \\ H_{\alpha_6} &= \Big(\varPhi\Big(\frac{1}{60}(H_2-H_3),0,0,0\Big),0,0,0,0,0\Big), \\ H_{\alpha_7} &= \Big(\varPhi\Big(\frac{1}{60}(H_1-H_2),0,0,0\Big),0,0,0,0,0\Big), \\ H_{\alpha_8} &= \Big(\varPhi\Big(0,0,0,\frac{1}{40}\Big),0,0,-\frac{1}{120},0,0\Big). \end{aligned}$
0902.0431_EQ0939	156	■ 0.471	\left.\begin{array}{l} \xrightarrow{\stackrel{f^{-1}}{ \longrightarrow}}\left(\begin{array}{ccc} h \left(b_{11}-c_{11}-\frac{\nu}{3}\right) & -h \left(c_{12}, b_{12}\right) & -h\left(c_{13}, b_{13} \right) \\ -h\left(\bar{b}_{12}, \bar{c}_{12}\right) & 0 & 0 \\ -h\left(\bar{b}_{13}, \bar{c}_{13}\right) & 0 & 0 \end{array}\right) \\ =\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & h\left(l_{33}\right) & -h\left(l_{23}, \bar{l}_{32}\right) \\ 0 & -h\left(l_{32}, \bar{l}_{23} \right) & h\left(l_{22}\right) \end{array}\right) \\ 0 \\ -h\left(\bar{l}_{11}\right) \end{array}\right) .	*(not rendered)*	$\xrightarrow{f^{-1}} \begin{pmatrix} \begin{pmatrix} h\Big(b_{11}-c_{11}-\frac{\nu}{3}\Big) & -h(c_{12},b_{12}) & -h(c_{13},b_{13}) \\ -h(\bar{b}_{12},\bar{c}_{12}) & 0 & 0 \\ -h(\bar{b}_{13},\bar{c}_{13}) & 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & h(l_{33}) & -h(l_{23},\bar{l}_{32}) \\ 0 & -h(l_{32},\bar{l}_{23}) & h(l_{22}) \end{pmatrix} \\ 0 \\ -h(\bar{l}_{11}) \end{pmatrix}$ $= \begin{pmatrix} \phi_{\boldsymbol{C}}(B,C)E_1+\frac{i\nu e_1}{3}E_1 \\ 2h(L)\times E_1 \\ 0 \\ -(\tau h(L),E_1) \end{pmatrix} = \varPhi(\phi_{\boldsymbol{C}}(B,C),h(L),-\tau h(L),-i\nu e_1)\begin{pmatrix} E_1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ1196	201	■ 0.477	`\begin{aligned} \mathfrak{g}_{01} & =\left\{(C, 0) \in \mathfrak{g}_{0} \mid C \in \mathfrak{s l}(5, C)\right\} \cong \mathfrak{s l}(5, C) \\ \mathfrak{g}_{02} & =\left\{(0, D) \in \mathfrak{g}_{0} \mid D \in \mathfrak{s l}(5, C)\right\} \cong \mathfrak{s l}(5, C) \\ \mathfrak{q} & =\mathfrak{g}_{1} \oplus \mathfrak{g}_{2} \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1} . \end{aligned}`	$\begin{aligned} \mathfrak{g}_{01} & =\left\{(C, 0) \in \mathfrak{g}_{0} \mid C \in \mathfrak{s l}(5, C)\right\} \cong \mathfrak{s l}(5, C) \\ \mathfrak{g}_{02} & =\left\{(0, D) \in \mathfrak{g}_{0} \mid D \in \mathfrak{s l}(5, C)\right\} \cong \mathfrak{s l}(5, C) \\ \mathfrak{q} & =\mathfrak{g}_{1} \oplus \mathfrak{g}_{2} \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1} . \end{aligned}$	$\begin{aligned} \mathfrak{g}_{01} & = \{(C,0) \in \mathfrak{g}_0 \mid C \in \mathfrak{sl}(5,C)\} \cong \mathfrak{sl}(5,C), \\ \mathfrak{g}_{02} & = \{(0,D) \in \mathfrak{g}_0 \mid D \in \mathfrak{sl}(5,C)\} \cong \mathfrak{sl}(5,C), \\ \mathfrak{q} & = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1}. \end{aligned}$
0902.0431_EQ1195	201	■ 0.485	`\begin{aligned} & (-*(y \wedge *(x \wedge a)), b)=-(x \wedge a, y \wedge b) \\ & \quad=\left(y, a_{1}\right)\left(x \wedge a_{2}, b\right)-\left(y, a_{2}\right)\left(x \wedge a_{1}, b\right)-(x, y)(a, b) . \end{aligned}`	$\begin{aligned} & (-*(y \wedge *(x \wedge a)), b)=-(x \wedge a, y \wedge b) \\ & \quad=\left(y, a_{1}\right)\left(x \wedge a_{2}, b\right)-\left(y, a_{2}\right)\left(x \wedge a_{1}, b\right)-(x, y)(a, b) . \end{aligned}$	$\begin{aligned} & (-*(\boldsymbol{y} \wedge *(\boldsymbol{x} \wedge \boldsymbol{a})), \boldsymbol{b}) = -(\boldsymbol{x} \wedge \boldsymbol{a}, \boldsymbol{y} \wedge \boldsymbol{b}) \\ & \quad = (\boldsymbol{y}, \boldsymbol{a}_1)(\boldsymbol{x} \wedge \boldsymbol{a}_2, \boldsymbol{b}) - (\boldsymbol{y}, \boldsymbol{a}_2)(\boldsymbol{x} \wedge \boldsymbol{a}_1, \boldsymbol{b}) - (\boldsymbol{x}, \boldsymbol{y})(\boldsymbol{a}, \boldsymbol{b}). \end{aligned}$
0902.0431_EQ1104	187	■ 0.490	`\varphi(R) R_{1}=\left[R, R_{1}\right], \quad R \in\left(\mathfrak{e}_{8(8)}\right)^{\tilde{\lambda} \gamma}, R_{1} \in\left(\mathfrak{e}_{8(8)}\right)_{-\widetilde{\lambda} \gamma} .`	$\varphi(R) R_{1}=\left[R, R_{1}\right], \quad R \in\left(\mathfrak{e}_{8(8)}\right)^{\tilde{\lambda} \gamma}, R_{1} \in\left(\mathfrak{e}_{8(8)}\right)_{-\widetilde{\lambda} \gamma} .$	$\varphi(R)R_1 = [R, R_1], \quad R \in (\mathfrak{e}_{8(8)})^{\widetilde{\lambda}\gamma}, R_1 \in (\mathfrak{e}_{8(8)})_{-\widetilde{\lambda}\gamma}.$
0902.0431_EQ1010	168	■ 0.494	`\begin{aligned} \mathfrak{e}_{8} & =\left\{R \in \mathfrak{e}_{8}{ }^{C} \mid \tau \tilde{\lambda} R=R \right\} \\ & =\left\{(\Phi, P,-\tau \lambda P, r, s,-\tau s) \in \mathfrak{e}_{8}{ }^{C} \mid \Phi \in \mathfrak{e}_{7}, P \in \mathfrak{P}^{C}, r \in i \boldsymbol{R}, s \in C\right\} . \end{aligned}`	$\begin{aligned} \mathfrak{e}_{8} & =\left\{R \in \mathfrak{e}_{8}{ }^{C} \mid \tau \tilde{\lambda} R=R \right\} \\ & =\left\{(\Phi, P,-\tau \lambda P, r, s,-\tau s) \in \mathfrak{e}_{8}{ }^{C} \mid \Phi \in \mathfrak{e}_{7}, P \in \mathfrak{P}^{C}, r \in i \boldsymbol{R}, s \in C\right\} . \end{aligned}$	$\begin{aligned} \mathfrak{e}_8 & = \{R \in \mathfrak{e}_8{}^C \mid \tau\widetilde{\lambda}R = R\} \\ & = \{(\varPhi, P, -\tau\lambda P, r, s, -\tau s) \in \mathfrak{e}_8{}^C \mid \varPhi \in \mathfrak{e}_7, P \in \mathfrak{P}^C, r \in i\boldsymbol{R}, s \in C\}. \end{aligned}$
0902.0431_EQ0952	158	■ 0.496	`\begin{aligned} D & =\left(\begin{array}{cc} B & L \\ -L^{*} & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right) \in \mathfrak{s u}(6), \\ \widetilde{M} & =\left(\begin{array}{cc} -N_{2}-N_{1} e_{1} & M_{2}+M_{1} e_{1} \\ M_{2}-M_{1} e_{1} & N_{2}-N_{1} e_{1} \end{array}\right), \quad M_{i}, N_{i} \in M(3, \boldsymbol{C}), \\ M & =M_{1}+i M_{2}, N=N_{1}+i N_{2} . \end{aligned}`	$\begin{aligned} D & =\left(\begin{array}{cc} B & L \\ -L^{*} & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right) \in \mathfrak{s u}(6), \\ \widetilde{M} & =\left(\begin{array}{cc} -N_{2}-N_{1} e_{1} & M_{2}+M_{1} e_{1} \\ M_{2}-M_{1} e_{1} & N_{2}-N_{1} e_{1} \end{array}\right), \quad M_{i}, N_{i} \in M(3, \boldsymbol{C}), \\ M & =M_{1}+i M_{2}, N=N_{1}+i N_{2} . \end{aligned}$	$\begin{aligned} & D = \begin{pmatrix} B & L \\ -L^* & C \end{pmatrix} + \frac{\nu}{3}\begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} \in \mathfrak{su}(6), \\ & \widetilde{M} = \begin{pmatrix} -N_2 - N_1e_1 & M_2 + M_1e_1 \\ M_2 - M_1e_1 & N_2 - N_1e_1 \end{pmatrix}, \quad M_i, N_i \in M(3, \boldsymbol{C}), \\ & M = M_1 + iM_2, N = N_1 + iN_2. \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value				